Are Stable Networks Stable? Experimental Evidence *

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July 2016

Abstract

We use a laboratory experiment to test the empirical content of different network stability concepts in a six-subject dynamic network formation game where link formation requires mutual consent. First, as predicted by theory, the game tends to converge to the pairwise-Nash stable network when it exists, and to remain in the closed cycle when no pairwise-Nash stable network exists. At the same time, stronger stability notions are more predictive of network outcomes. Second, the analysis of single decisions indicates that myopic rationality is predominant, but that we also observe interesting systematic deviations from it. In particular, deviations are more frequent when actions are more easily reversible and when they involve smaller marginal losses. Third, we also notice a significant heterogeneity in behavior across subjects, ranging from extreme myopic rationality to choices close to random.

Keywords: social networks, stable networks, myopic rationality, laboratory experiments.
JEL Classification: C73, C92, D85.

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*This paper is a substantial revision of an earlier draft entitled "The Strategic Formation of Networks: Experimental Evidence". We would like to thank Marina Agranov, Douglas Bernheim, John Beshears, Manuel Castro, Gary Charness, Catherine Hafer, Matthew Jackson, Sera Linardi, Muriel Niederle, Tom Palfrey, Charlie Plott, Saurabh Singhal, Bob Slonim, Charlie Sprenger, Leeat Yariv, and the audiences at Caltech, USC, University of Utah, UC Santa Barbara, and Stanford for helpful comments. We also gratefully acknowledge the financial support of the Microsoft Corporation (Carrillo), and the American-Indonesian Cultural and Educational Foundation (Gaduh).

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1 Introduction

Social networks shape a variety of social and economic interactions and their importance has been increasingly recognized in economics (Jackson, 2014). Among economists, a key question of interest is on how incentives shape networks that are formed by self-interested agents in a decentralized fashion. To put it in another way, given incentives, we would like to predict the structure of the *stable networks* that will emerge among self-interested agents. Studies of stable forms of cooperation has a deep root in economics, particularly in the context of decentralized matching (Gale and Shapley, 1962). More recently, theorists tackle a similar question in the context of network formation with the introduction of solution concepts that predict the kind of stable networks that will emerge under different assumptions about individual behaviors.

One of the stability notions most commonly used in the literature is *pairwise stability*. It formally describes the networks that emerge when link formations among pairs of self-interested individuals require mutual consent but link deletion can be performed unilaterally (Jackson and Wolinsky, 1996). Pairwise stability is a relatively weak notion, as it is only robust to pairwise deviations (Jackson, 2008). Further refinements lead to strongly-stable solution concepts that require stable networks to be robust to (coordinated) deviations by any number of agents (Dutta and Mutuswami, 1997; Jackson and van den Nouweland, 2005). Meanwhile, the literature has also analyzed the network evolution process. Jackson and Watts (2002) study the dynamics of social networks when the actions of all subjects are *myopic rational* – i.e., each link is decided solely based on its current net benefit without strategically thinking ahead. The authors show that under such myopic decision rules, the system always converges to a pairwise-stable network if it exists.

Despite significant theoretical advances, there is a very small literature examining pairwise stability in dynamic linking games with mutual consent (Pantz, 2006; Kirchsteiger et al., 2013) and no work has examined the empirical robustness of the different stability notions.[1] Empirical evidence on robustness will help researchers narrow down the list of stability notions from the variety of theoretical (equilibrium) constructs in network formation. Since testing stability using observational data is difficult,[2] we build a controlled laboratory experiment with the following three objectives: (i) to examine how well different stability notions predict the outcome of network formation games; (ii) to provide novel experimental evidence on correlates of deviations from myopic rational decisions; and (iii)

[1] In contrast, empirical analyses of stability in decentralized matching settings have received more attention (see, e.g., Chen and Sönmez, 2006; Echenique and Yariv, 2013; Echenique et al., forthcoming).

[2] Methodological papers that analyze peer effects in endogenously-formed social networks have *assumed* that the observed networks are pairwise stable (see, e.g., Boucher and Mourifié, 2013; Sheng, 2014).

to investigate heterogeneity across subjects on the level of myopic rational choice and the reasons for deviating from it.

Our experimental game slightly modifies the dynamic linking game introduced by Watts (2001). Each game is played with six subjects. This introduces significant complexity in the network formation process but also allows a rich game structure, where we can vary the existence and type of stable networks, and also avoid focal networks.

We study the data in three different ways: network outcomes, single decisions and choices of subjects. From the analysis of final networks, we find that stability is an empirically meaningful notion that can predict network outcomes reasonably well. In the absence of a pairwise Nash stable (PNS) network, the game does not converge but stays within a closed cycle as the theory predicts. When a unique PNS network exists, we find evidence of convergence to it. Importantly, however, the likelihood of convergence depends on how strongly stable the unique PNS network is. As developed in Jackson and van den Nouweland (2005), stability strength is related to the asymmetry of payoffs among subjects (Result 1).

Our analysis of single decisions empirically qualifies a key behavioral assumption of Jackson and Watts (2002)'s theory of social network evolution. Their predictions rely on having agents that (almost) always make myopic rational choices. Although a high percentage of decisions in our experiment are myopic rational (between 76 percent and 97 percent depending on the turn and treatment), we also find evidence of *systematic* deviations. We build an empirical model to test for correlates of deviations from myopic rationality and find three intuitive situations that affect the probability of deviations. First, deviations are more common in early turns. This is natural because, in our design, subjects are guaranteed 12 turns before they enter a probabilistic match-ending phase. Second, deviations are more frequent when they imply keeping an excessive number of links, presumably because future link removals are easy in the sense that they do not require consent of the other subject. These two deviations suggest that subjects are more likely to "experiment" with decisions that are not myopic rational if they feel that such actions can be reversed later on. Third, subjects deviate more often when the marginal payoff losses are small. This is consistent with a theory of "imperfect choice", where mistakes are inversely related to their cost (Result 2).

A cluster analysis performed at the subject level reveals substantial heterogeneity in behavior across individuals. About 25 percent of our subjects follow remarkably closely the myopic rationality precepts of Jackson and Watts (2002) in all turns and treatments. These subjects obtain the highest earnings. The next 40 percent of subjects are a slightly less consistent version of these subjects, with somewhat lower levels of myopic rationality

and small variations across turns. On the other extreme, 10 percent of subjects are lost. They deviate from myopic rationality frequently and with no discernible pattern. They obtain the lowest earning. The remaining 25 percent of subjects exhibit an interesting strategic behavior characterized by low myopic rationality at the beginning of the game and when the marginal cost of doing so is low. Myopic rationality increases dramatically in late turns (by around 20 percentage points or p.p.) when their choices are more likely to be irreversible (Result 3).

Our paper contributes to the growing number of experimental studies on network formation.[3] The bulk of the literature focuses on examining stability in the unilateral link formation framework of Bala and Goyal (2000) or the bilateral link announcements game of Myerson (1991).[4] Closer to our setting is the (smaller) experimental literature on dynamic linking games with mutual consent. Pantz (2006) and Kirchsteiger et al. (2013) examine outcome selection given multiple PNS networks, and find evidence of (limited) farsighted rational behavior. Of these, the experiment of Kirchsteiger et al. (2013) also implements the dynamic linking model of Watts (2001). However, our focus is different in that we design our experiment to systematically study the predictive power of different stability notions and the incentives to deviate from myopic rational decisions in the absence of multiple equilibria.[5]

The paper is organized as follows. In Section 2, we present the conceptual framework and the theoretical literature pertinent to our experiment. Section 3 describes the experimental design and introduces our treatments. Then, in the following three sections, we present our analysis at the final network level (Section 4), single decision level (Section 5) and subject level (Section 6). Section 7 concludes.

2 Network environment and basic definitions

A network is a collection of links that connect a set of agents. A link between two agents forms if and only if both decide that it is worth forming. Each link is costly for both agents and this cost is non-transferable. Meanwhile, the benefit depends on and is a

[3] There is also a related experimental literature on equilibrium selection in network games (see e.g. Fatas et al., 2010; Charness et al., 2014).

[4] See e.g., Callander and Plott (2005), Berninghaus et al. (2006), Berninghaus et al. (2007), Falk and Kosfeld (2012) and Goeree et al. (2009) for the first line of investigation and Conte et al. (2015), Di Cagno and Sciubba (2008) and Burger and Buskens (2009) for the second one.

[5] More recently, some studies propose interesting variants of the bilateral linking game, where payoffs are received at each turn (Teteryatnikova and Tremewan, 2015), payoffs are pair-specific (Comola and Fafchamps, 2015), subjects have limited observation of the network structure (Caldara and McBride, 2015), or networks face threats of disruption (Candelo et al., 2014).

strictly increasing function of the size of the network component that an agent belongs to. We distinguish between a *network* and a *component*. A network describes the link configurations that include the full graph (all agents) while a component is a sub-graph in which there exists a path linking any two agents. In our setup, all agents in a component receive the same benefit. Payoffs are computed as the difference between benefits and costs.

A number of theoretical approaches analyze endogenous network formation among rational, self-interested agents when link formation requires mutual consent (see Bloch and Jackson, 2006). Myerson (1991) explicitly considered a linking game and used its Nash equilibrium to define the stable networks. In this game, all agents simultaneously list all other agents that each agent wants to link with. Agent i's strategy set is an n-tuple of binary variables indicating his willingness to link with each of the agents in the game. A link between i and j forms if both agents mutually agree to link. A strategy profile is a Nash equilibrium of the game if and only if no subject can benefit from any unilateral deviation from the strategy. A network is *Nash stable* if it is induced by a (pure strategy) Nash equilibrium of the linking game.

Nash stability does not allow some agents to coordinate to improve their payoffs. Jackson and Wolinsky (1996) relaxed this restriction with their notion of pairwise stability. Pairwise stability allows for pairwise deviations and rules out networks that are "intuitively unstable" when formed by strategic actors. A network is *Pairwise stable* if: (i) all existing links are weakly preferred by both agents in the link and are strictly preferred by at least one of them; and (ii) all non-existing links are such that at least one of the agents on the non-existing link strictly prefers its absence. Pairwise Nash stability is a further refinement that combines both notions: a network is *Pairwise Nash stable* (PNS) if and only if it is both Nash and pairwise stable (Bloch and Jackson, 2006).[6]

While robust to pairwise deviations, a PNS network is not necessarily robust to deviations by a coalition of (more than two) agents. Dutta and Mutuswami (1997) introduced a stronger notion that attempted to account for such deviations. A vector of strategies constitute a strong Nash equilibrium of the linking game if and only if no coalition can deviate in such a way that each member of the coalition is strictly better off. The network induced by such strategies is referred to as *Strongly stable* in the sense of Dutta and Mutuswami (SSDM). Jackson and van den Nouweland (2005) further refined that concept by allowing coalition-wise deviations that made some members strictly better-off and others only weakly better-off. We refer to the network induced by such strategies as *Strongly*

[6]Nash stability and pairwise stability are two distinct solutions. Bloch and Jackson (2006, Remark 1) show that there exists profiles of payoff structures of the linking game such that sets of Nash stable and pairwise-stable networks do not intersect, even though neither set is empty.

stable in the sense of Jackson and van den Nouweland (SSJN). To sum up, we consider three definitions of network stability, PNS, SSDM and SSJN, that differ in how robust they each are to coalition deviations. By construction, SSJN $\subseteq$ SSDM $\subseteq$ PNS: the set of SSJN networks is a subset of the set of SSDM networks which itself is a subset of the set of PNS networks.

In a dynamic linking game where pairs of agents randomly meet and make linking decisions, Jackson and Watts (2002) show how pairwise stability can help predict the network outcome. Suppose that each linking action is *myopic rational* – to wit, it optimizes the marginal payoff from the link under consideration (and not on the option value of forming or severing links in the future). Hence, a link forms if both agents are weakly better-off with it and at least one is strictly better-off. Conversely, an existing link breaks if at least one agent is strictly better-off without it. If all actions are myopic rational, then the network evolves following an improving path. Starting from any network, Jackson and Watts (2002, Lemma 1) show that improving paths lead to either a pairwise-stable network or, when none exists, a *closed cycle*. A set of networks forms a closed cycle if no network in the set is on the improving path of a network that is not in the set.

Beyond stability, we also include as a benchmark the set of networks that arises when agents maximize the sum of payoffs received by all agents. Following Jackson and Wolinsky (1996), we refer to such networks as *efficient*.

3 Experimental setting and procedures

3.1 The basic configuration

We are interested in environments with a *large number of network configurations* where links are costly and mutual pairwise consent is needed to form a link but not to break it. To this end, we implement a stochastic dynamic linking game that slightly modifies the procedure proposed by Jackson and Watts (2002). We consider networks with $n = 6$ subjects. This implies $n(n-1)/2 = 15$ possible bilateral undirected links between different subjects, and therefore $2^{n(n-1)/2} = 32,768$ possible networks. Notice that many networks differ from each other only on the identity of subjects in the different positions. We say that two networks have the same *network structure* if they are identical up to a permutation of the identity of subjects.

We consider a particular payoff structure. Every subject in a component receives the same benefit, which is a strictly increasing function of its size, while the cost of links is

borne solely by their owners.[7] This design choice serves two objectives. First, we want to maintain tractability given the large number of possible networks. This payoff structure limits the set of stable and efficient networks to be a subset of *minimally-connected* networks (a network is minimally-connected if the removal of any existing link increases the number of components, see Bala and Goyal, 2000). When the benefit is strictly a function of the component size, removing a link that does not reduce the component size is always Pareto improving. With 6 subjects, there are 20 minimally-connected network structures. Second, we also want to simplify the game enough to minimize the likelihood of participants' computing mistakes. As such, the allocation of benefits deviates from the usual connections model where links have indirect benefits that decay with distance at a rate δ (Jackson and Wolinsky, 1996). Formally, we set $\delta = 1$. Finally, we simplify further by maintaining a constant unit cost of a direct link both within and across treatments.

3.2 Experimental design

Each match consists of multiple turns and starts with an empty network. At each turn, the computer randomly pairs the six subjects. Subjects then choose their actions with respect to their partner in the pair. A new turn begins after all subjects have taken their actions. If all subjects are satisfied with the network outcome, they can collectively end the game. We implement a match-ending rule that provides enough opportunities for subjects to converge but, at the same time, allows decisions to be meaningful and the experiment to be time manageable. Subjects play for 12 turns unless all subjects are satisfied with the network. Afterwards, each turn is the last one with probability $p = 0.2$, providing an additional $1/p = 5$ turns on average. With this probabilistic match-end rule, we hope to mitigate the last-round effects. More importantly, it allows for an interesting comparison of behavior before and after Turn 12. Finally, notice that each turn is composed of six decisions, one for each subject, providing a fairly large number of individual decisions per match ($17 \times 6 = 102$ on average, unless subjects decide to stop before).

Figure 1 shows the user interface. At each turn, subjects make decisions by clicking on one of the action buttons displayed on the lower left section of the screen. If a subject is not linked to his partner, he chooses whether to "Propose" a link or "Pass Turn". If he is linked, he chooses whether to "Remove" a link or "Pass Turn". Once a pair of partners

[7] An application of this setup is for studying the endogenous formation of risk-sharing networks. Bramoullé and Kranton (2007) shows that with repeated interactions, an arrangement in which individuals commit to share monetary holdings equally with linked partners amounts to equal sharing within network components. Another example would be the case of club goods (such as those provided by religious or social groups, see e.g., Berman, 2000) without centralized coordination. In this setup, all members benefit from having an additional member, but participation requires individuals to maintain costly direct links.

have taken their actions, the result is immediately displayed on the screen. Hence, when each subject makes his decision, he observes the latest state of the network. Showing the latest network configuration *within a turn* allows us to cleanly determine whether each individual decision reflects a myopic rational behavior or otherwise.[8] If a subject is not only satisfied with the relationship with his partner but also with the overall network, instead of choosing "Pass Turn", he can choose "Network OK".[9] As mentioned above, the match immediately ends if all subjects within a turn choose "Network OK".

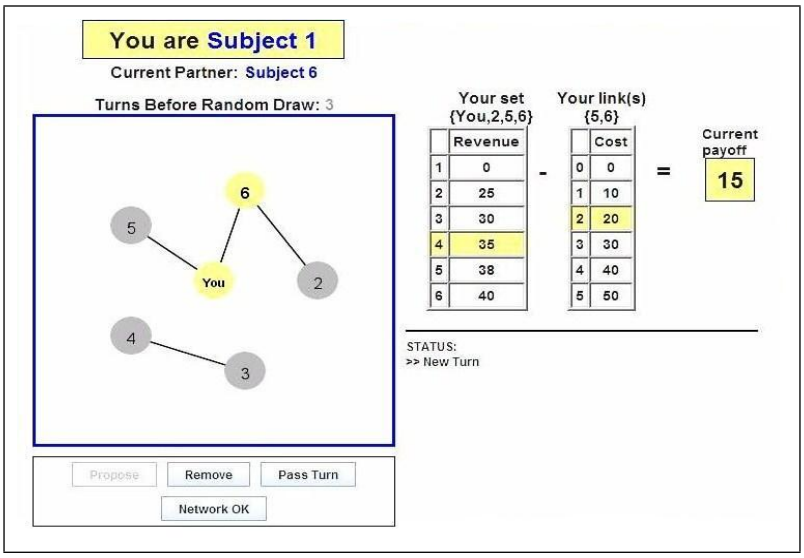


Figure 1: User interface for the linking game.

The user interface displays on the left side of the screen all the pertinent information: the subject's role, the role of the person he is currently matched with, whether the current turn is a potential terminal turn and, naturally, the current network configuration. It displays on the right side of the screen the benefit of the subject as a function of the size of the component he is in, the cost as a function of his number of direct links, and his net payoff given the current network configuration. This succinct but comprehensive visual display allows the subject to compute rather easily not only the net value of adding or removing an existing link (i.e., the improving path) but also his payoff in any other network configuration. Finally but crucially, notice that the node representing the subject is always

[8] On the other hand, it could encourage a war of attrition, where subjects wait to see what others do in a turn before choosing their own action. There is no evidence in our data of such behavior.

[9] Once a subject chooses "Network OK", he does not need to choose further actions until the network changes. To avoid mistakes, all of his action buttons become inactive. These buttons are immediately reactivated following a change in the network.

located at the center and labeled "You". The nodes representing the other subjects in a match are labeled by their roles and surround the subject's node in clockwise order at an equal distance from it. By always putting the subject's node at the center, even though the underlying connections between subjects in a match are identical, each subject sees a different graphical representation. We therefore avoid leading participants towards focal networks such as the star or wheel network.

3.3 Treatments

The experiment consists of three main treatments. Figures 4, 5 and 6 illustrate how we design them. First, we construct a "supernetwork" that contains all 20 minimally-connected six-node network structures (labeled $\{A\}$ to $\{T\}$). Then, we add all the arcs connecting pairs of networks that differ from one another by a single link. Network structures are ordered from top to bottom based on their number of links and those with identical number of links are lined up on the same row. Each network is therefore connected by an arc to one or more networks in the row above and one or more networks in the row below it. The direction of the arc represents the improving path: forming a new link (arrow pointing to the row below) or removing an existing link (arrow pointing to the row above).[10]

Naturally, the improving path depends on the payoffs of each treatment. Contrary to some existing network experiments, we construct payoff functions that do not follow any simple functional form. Instead, our payoffs obey two simple restrictions: the benefits are strictly increasing in component size and the unit cost of a link is constant. This flexibility simplifies the task of choosing payoffs that can support either no PNS network or a unique PNS network that is different from the empty or full network and may or may not be robust to coalition deviations.

Table 1 summarizes the key information of the three treatments, namely payoff functions (benefits and costs) and outcome predictions. Treatment **N**, has *no* PNS network and a closed cycle comprising networks $\{B, C, D, F, G, H, N\}$ (the shaded area). Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$ (for single) have each a *unique* PNS network, $\{L\}$ and $\{H\}$ respectively, which differ in their robustness to coalition deviations. In Treatment $\mathbf{S}^-$, the PNS network $\{L\}$ is SSDM. In this network, no coalition can deviate to another network where all members are *strictly* better off. However, there exists a coalition of three agents – the two agents who are in the middle of the three-node components and one other agent – whose deviation can lead to another stable network with the same structure $\{L\}$, where all three

[10] Networks that are not minimally connected are necessarily off the improving paths. They are omitted unless a match ends in one of them.

agents are weakly better off and one of them is strictly better off. Meanwhile, the PNS network $\{H\}$ in Treatment $\mathbf{S}^+$ is SSDM and SSJN since no such coalition exists. Network $\{H\}$ in Treatment $\mathbf{S}^+$ is therefore more robust to coalition deviations than network $\{L\}$ in Treatment $\mathbf{S}^-$.

To facilitate the comparison of final outcomes across treatments, subjects always start in the empty network $\{A\}$ and the efficient networks are always the same: all of the minimally-connected full networks, that is, $\{O, P, Q, R, S, T\}$. We picked multiple efficient networks with a focal one ($\{T\}$, the line that comprises all agents) to give a fair chance to the efficient outcome.

Table 1: Efficient and stable networks

Treatment	Benefit (size of component)						Link Cost	PNS	SSDM	SSJN	Efficient
	1	2	3	4	5	6					
N	0	20	30	39	42	43	15	None	None	None	[6]*
$\mathbf{S}^-$	0	19	36	42	44	45	15	$[3\text{-}3]_{\{L\}}$	$[3\text{-}3]_{\{L\}}$	None	[6]*
$\mathbf{S}^+$	0	29	36	41	43	44	15	$[2\text{-}2\text{-}2]_{\{H\}}$	$[2\text{-}2\text{-}2]_{\{H\}}$	$[2\text{-}2\text{-}2]_{\{H\}}$	[6]*

Note: The numbers in brackets refer to the size of each component. *Includes $\{O, P, Q, R, S, T\}$.

Finally, we also considered a fourth treatment with *multiple* PNS networks, Treatment **M**. Due to a computation error during programming, instead of the intended two stable networks (one SSDM and one SSJN), our configuration had two pairwise-stable networks, one PNS network and one SSDM network. This complicates the data analysis and pollutes the comparisons with the other treatments. We therefore decided to concentrate on Treatments **N**, $\mathbf{S}^-$, and $\mathbf{S}^+$ and relegate a brief analysis of Treatment **M** to Appendix A.

3.4 Implementation

The experiment was conducted in the California Social Science Experimental Laboratory at UCLA. All experimental subjects were UCLA students. We conducted 8 experimental sessions with 12 subjects in each session. With 12 subjects, there are always 2 groups of 6 subjects in each session, playing simultaneously 2 matches. Each subject played each of the four treatments twice. We shuffled the order of the treatments to neutralize the possible effects from the ordering within a session.[11] The analysis in the main text utilizes a total

[11] Specifically, the order of the treatments is such that: (i) the orders of the treatments in the first half and the second half of each session were different; (ii) no two sessions have identical treatment sequences; and

of 96 match observations (32 matches of treatments $\mathbf{N}$, $\mathbf{S}^-$, and $\mathbf{S}^+$) from 96 subjects (12 subjects in 8 sessions). The analysis in the appendix utilizes 32 match observations (the remaining treatment) from the same subjects.

To introduce anonymity in game play, after each match we reshuffled subjects into new groups and assigned a new role (1 to 6) to each subject. Each session lasted between 90 and 120 minutes. No subject took part in more than one session. Participants interacted exclusively through computer terminals without knowing the identities of the subjects they played with. Before the paid matches, instructions were read aloud and two practice matches were played to familiarize participants with the computer interface and procedure. After that, participants had to complete a quiz to ensure they understood the rules of the experiment.

At the end of each match, subjects obtained a payoff based on the size of the component they were in (benefit) and the number of direct links (cost). Participants were endowed with experimental tokens and they could earn or lose tokens. At the end of the session, the payoffs in tokens accumulated from all experimental games were converted into cash, at the exchange rate of 4 tokens = \$1. Participants received a show-up fee of \$5, plus the amount they accumulated during the paid matches. Payments were made in cash and in private. Matches lasted between 13 and 36 turns, with an average of 16.8 turns. There was a significant spread in winnings: including the show-up fee, participants earned between \$11 and \$43 with an average of \$29. A copy of the instructions is included in Appendix C.

4 Network outcomes

This section reports the results on *network outcomes*. First, we show evidence on the predictions that link pairwise stability to the outcome of the dynamic linking game that were derived by Jackson and Watts (2002). Next, we compare the predictive powers of unique PNS networks whose stability differ in their "strength". The results of our analysis are based on the final network outcomes and those conditional on convergence. We employ the operational definition of convergence suggested by Callander and Plott (2005), as the lack of change in the final $T = 3$ turns.[12]

Before describing the main results, we first study whether our subjects understand the basic tenets of the game. Table 2 shows that subjects consistently avoid networks that are not minimally connected. As discussed above, removing a link that does not reduce the

(iii) each treatment was implemented in exactly two (out of eight) sessions for each order in the sequence.

[12] $T = 3$ is arbitrary. It corresponds to 18 individual decisions which seems reasonably large. With a larger T, convergence decreases but the qualitative conclusions are very similar ($T = 5$ is not presented for brevity but it is available from the authors).

component size strictly increases the payoffs of both agents involved, without affecting any other. The second column in Table 2 suggests that subjects understand this principle, as only 4 out of 96 matches end up in a network that is not minimally connected.

Table 2: Network outcomes

Treatment	Not Min. Conn.	PNS	Closed cycle	Efficient	N
	Panel A. **All**				
N	2 (0.06)	—	21 (0.66)	3 (0.09)	32
$\mathbf{S}^-$	2 (0.06)	10 (0.31)	—	2 (0.06)	32
$\mathbf{S}^+$	0 (0)	15 (0.47)	—	0 (0)	32
	Panel B. **Conditional on Convergence**				
N	1 (0.20)	—	1 (0.20)	2 (0.40)	5
$\mathbf{S}^-$	0 (0)	4 (0.27)	—	1 (0.07)	15
$\mathbf{S}^+$	0 (0)	9 (0.56)	—	0 (0)	16

Notes: Share with respect to N in parentheses. Convergence defined as having no change in the final 3 turns (= 18 individual decisions).

4.1 Network convergence, pairwise stability and strong-stability

Hypothesis 1 *The dynamic link formation process remains in the closed cycle when no PNS network exists and leads to the PNS network when it exists. In that case, the stronger strong-stability notion (SSJN) better predicts convergence than the weaker notion (SSDM).*

According to the theory briefly presented in Section 2, when agents follow improving paths, the linking game will lead to either a PNS network or, when a PNS network does not exist, a network in the closed cycle. In the latter case, we should not observe convergence. However, different stability concepts are more or less robust to different types of coordinated deviations. Therefore, it is reasonable to think that networks which are not only stable but

also robust to all sorts of coalitional deviations are more likely to be empirically observed than those which are robust only to a limited type of deviations.

With this idea in mind, our first result consists in an analysis of the realized network outcomes.

Result 1 *In Treatment* $\boldsymbol{N}$*, games tend to end in a network within the closed cycle. In Treatments* $\boldsymbol{S}^-$ *and* $\boldsymbol{S}^+$*, games tend to end in the PNS network. However, convergence to the PNS network is observed significantly more frequently in* $\boldsymbol{S}^+$ *than in* $\boldsymbol{S}^-$*.*

Our results provide evidence in favor of Hypothesis 1. Table 2 summarizes how well the different types of network structures predict the final network. In the absence of a PNS network (Treatment $\mathbf{N}$), 66 percent of the matches end in a network within the closed cycle.[13] In treatments with a single PNS network ($\mathbf{S}^-$ and $\mathbf{S}^+$), subjects end up in the stable network 31 percent and 47 percent of the time, respectively. In addition, matches rarely end in an efficient network.

As predicted by theory, convergence is least frequent in Treatment $\mathbf{N}$ when no PNS exists (5 out of 32 matches) compared to about half the network outcomes for Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$. Indeed, in Treatment $\mathbf{N}$, myopic rational subjects are expected to move indefinitely within the closed cycle and along the improving path. Figures 4, 5, and 6 display the number of final outcomes in each network both conditional on no change in the final 3 turns (labeled C) and unconditional on convergence (labeled U).

In Appendix B.1 we provide a complementary analysis based on the *distance* between the observed and predicted (efficient, PNS, closed cycle) outcomes, both conditional and unconditional on convergence. The conclusions are similar. The distance is smaller between the observed outcome and the networks in the closed cycle (for Treatment $\mathbf{N}$) and the PNS network (for Treatment $\mathbf{S}^+$) than between the observed outcome and any of the efficient networks. However, the result is less clear cut for Treatment $\mathbf{S}^-$.

Finally, we find that the final network is more likely to be PNS in Treatment $\mathbf{S}^+$ relative to $\mathbf{S}^-$, especially when focus on convergent networks. As mentioned earlier, unconditional on convergence, the share of final networks that are PNS is larger in Treatment $\mathbf{S}^+$ (47 percent) compared to $\mathbf{S}^-$ (31 percent), even though this 16 p.p. difference is not statistically significant at conventional levels.[14] When we restrict the sample to convergent networks,

[13] Matches stay within the closed cycle most of the time: in 56 percent of all the moves after Turn 6 subjects remain in a network within the closed cycle. The result should not be overemphasized as the closed cycle is rather large (7 out of 20 network structures).

[14] A simple one-tailed *t-test* (*z-test*) of the likelihood of convergence to the PNS network between Treatments $\mathbf{S}^+$ and $\mathbf{S}^-$ yields a *p-value* of 0.104 (0.100).

however, this difference increases to $56 - 27 = 29$ p.p. (Panel B of Table 2) and is statistically significant.[15] This difference in the likelihood of convergence to the PNS between Treatments $\mathbf{S}^+$ and $\mathbf{S}^-$ is intriguing. Our result suggests that the disparity may be rooted in the differences in strength of the stability between network $\{H\}$ in Treatment $\mathbf{S}^+$ and network $\{L\}$ in Treatment $\mathbf{S}^-$. While the former is robust to coalition deviations by a set of agents where some are strictly better-off and others weakly better-off (SSJN), the latter is only robust to coalition deviations by agents who are all strictly better-off (SSDM).

The difference between these two stability concepts may seem minor, but it is not. Indeed, a property of any (non-connected) SSJN network is that the value of each component is equally distributed among its members. This is called *component-wise egalitarian allocation rule* (Jackson and van den Nouweland, 2005), and it is not shared by SSDM networks. In our game, it means that the payoff symmetry of the [2-2-2] network $\{H\}$ in Treatment $\mathbf{S}^+$ ensures that, once it is reached, there is simply no room for improvement for any subject. By contrast, the [3-3] network $\{L\}$ in Treatment $\mathbf{S}^-$ is only SSDM and therefore does not satisfy this property. In particular, the two agents at the center of each component have strong incentives to deviate, in the hope of reaching later on the same configuration but with someone else bearing the cost of two links.[16]

4.2 Payoffs

Table 3 presents the mean payoffs obtained by subjects in each treatment. These values are compared with the average of the mean payoffs had they ended in a network in the closed cycle (Treatment $\mathbf{N}$) and in the unique PNS network (Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$). We also compare them to the payoffs in the efficient networks. We find significant losses relative to the payoffs that could be collectively obtained: empirical payoffs are between 46 percent and 71 percent of the payoffs generated by the efficient networks. It means that playing non-cooperatively comes at a substantial cost. In fact, the observed payoffs are smaller but close to the payoffs in the unique PNS network for Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$. The results are similar when we consider only the empirical payoffs of the convergent networks. Finally, the payoffs for Treatment $\mathbf{N}$ are 50 percent higher than the average payoffs in the closed

[15] Assuming observational independence, a one-tailed *t-test* of the difference in the mean shares of the PNS final network among convergent networks has a *p-value* of 0.051. Since participants are reshuffled between matches within a session, this independence assumption may not hold. Using a *t-test* with standard errors that are clustered at the session level, we obtain a *p-value* of 0.095. Obviously, these effects are imprecisely estimated in part due to the the small number of match-level observations.

[16] The only case where subjects in an SSJN network do not obtain the same payoff is when the SSJN network is fully connected (Jackson and van den Nouweland, 2005). The reason why such network is still SSJN is simply that there is no other component that subjects can join to improve their payoff.

cycle. Therefore, subjects not only remain mostly within the close cycle, they even stay more often in the high payoff networks within the cycle.[17]

Table 3: Summary of average per-capita payoffs

Treatment	Empirical		Predicted		
	All	Convergent	PNS	Closed cycle	Efficient
N	63.1 (26.6)	—	—	43.1 †	108
$\mathbf{S}^-$	85.4 (17.9)	84.8 (16.5)	96	—	120
$\mathbf{S}^+$	81.5 (12.3)	79.4 (12.8)	84	—	114

Standard deviation in parenthesis. †Unweighted average payoffs of all networks in the cycle.

4.3 Summary

Subjects in our game understand the strategic nature of network formation and avoid non-minimally connected components. The collective gain (network efficiency) does not appear to be an important motivation in any treatment. As predicted by theory, in the absence of a PNS network, the final network remains in the closed cycle whereas in the presence of a PNS network, it tends to converge to it. Interestingly, networks that are more robust to coalition-wise deviations predict convergence more accurately (SSJN better than SSDM).

5 Single decisions

In this section, we estimate an empirical model to understand and predict each decision to stay or stray from the improving paths. Due to the uncertainty from the random pairing of partners, it is difficult to calculate an *ex ante* optimal strategy in this game. The least risky response to this uncertainty is to always take myopically rational actions. In our games, this strategy would lead to the close cycle in Treatment **N** and to the PNS network in Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$. However, it may not necessarily yield the maximum possible *individual* outcome from the stable network, unless the stable network follows the component-wise egalitarian allocation rule (as in SSJN in Treatment $\mathbf{S}^+$). Treatment $\mathbf{S}^-$

[17] In particular, almost half of the final outcomes within the close cycle are in networks $\{F\}$ and $\{N\}$, the two highest paying networks in the cycle (with payoffs of 66 and 76, respectively).

is an example where the maximum individual outcome requires subjects not only to reach the SSDM, but also be at one end of the network component by the end of the match.

Such a motivation among some subjects in a game may lead to alternative strategies that do not consist solely of myopic rational actions. We consider two intuitive features of link formation that might influence these strategies: the end-game rule and the asymmetry between link formation and link deletion. With regards to the former, with 12 guaranteed turns, subjects may be more willing to play riskier strategies earlier in the game. Meanwhile, the latter may motivate subjects to deviate from myopic rationality mainly by accumulating extra links, since link deletion can be implemented unilaterally whereas link formation requires mutual consent. Lastly, the willingness to experiment by straying from the improving paths may also be influenced by the potential loss from that particular deviation.

5.1 Descriptive statistics

At each turn, each subject in a pair must choose to either "act" or "pass". If subjects in the pair are initially unlinked, acting implies proposing a link and passing implies remaining unlinked. If subjects are initially linked instead, acting implies removing a link and passing implies remaining linked. We are interested in the extent to which actions are myopic rational in each of these four cases and for each treatment.

Table 4 summarizes the proportion of myopic rational actions across turns. We organize the data into four groups of turns. We use the last certain turn that subjects get unless everyone agrees on the network outcome (Turn 12) as a natural point to partition and further split each of these partitions into two. This split captures behaviors at different stages. First, subjects attempt to get familiar with the current match and try different strategies which, with high probability, can be reversed later if desired (Turns [1-6]). Then, subjects adjust their behavior as the last certain turn approaches (Turns [7-12]). After that, subjects enter the random stopping phase where, presumably, they behave under the assumption that matches can be terminated at any time (Turns [13-18]). Finally, Turns 19 and above are set in a another category because the sample size is dramatically reduced as turns advance and the sample becomes non-representative of the population.[18]

Although formal tests are presented in the regression analysis of Section 5.2, Table 4 is instructive. Panel A suggests a statistically significant upward jump in myopic rationality when subjects enter the probabilistic match-ending phase (between [7-12] and [13-18]).

[18] The split is arbitrary. Similar results are obtained if the first and third partition are changed marginally. The key issue is to introduce a separation at Turn 12.

Panel B investigates myopic rationality by type of decision. We examine choices under four mutually exclusive conditions, namely when the rational action is: (i) to pass and stay unlinked; (ii) to pass and stay linked; (iii) to remove an existing link; and (iv) to propose a new link. By comparing conditions (i) with (ii), and (iii) with (iv), we find evidence that individuals deviate more from improving paths in decisions that reduce the number of links than in decisions that increase the number of links. By comparing conditions (i) with (iii), and (ii) with (iv), subjects deviate more by being "over-passive" (failing to act when they should) than "over-active" (acting when they should not).[19] However, our regressions below suggest that this last result does not hold once we control for subject fixed effects and the marginal payoff from myopic rational choices. Panel C displays myopic rationality across treatments and confirms the results of panel A: in all three treatments, subjects are significantly less myopic rational before Turn 12 than after Turn 12 at the 0.1 percent level. A graphical illustration of the results in panels B and C is presented for every turn of the game (up to Turn 18) in Appendix B.2.

Finally, Table 5 presents the number of instances in which subjects choose to "enter", "stay" and "leave" the PNS network, broken down by treatment and turn. The last column reports the total number of turns in that set. We notice that subjects are prone to leave the PNS network in Turns [1-12], but this tendency is dramatically reduced when each turn can be final, especially for Treatment $\mathbf{S}^{+}$.

5.2 Empirical framework

5.2.1 Specification

As a formal test, we estimate a linear probability model (LPM) with individual fixed effects and regress the probability that a subject chooses the myopic rational action on the attributes of the problem.[20] For each treatment, we estimate the following specification:

$$\mathbb{P}(Y_{nt}^{ij} = 1 \mid \mathbf{X_{nt}^{ij}}, c_n) = \beta_0 + \mathbf{X_{nt}^{ij}}\,\boldsymbol{\beta} + c_n \tag{1}$$

where Y_{nt}^{ij} indicates whether the action that moves subject n from network i to network j in the supernetwork at Turn t is myopic rational ($= 1$) or not ($= 0$), and $\mathbf{X_{nt}^{ij}}$ captures the

[19] A set of t-tests (not reported for brevity) confirms that for each turn group, the mean differences in myopic rationality both between conditions (i) and (iii) and between conditions (ii) and (iv) are negative and statistically significant at the 0.1 percent level.

[20] We choose LPM with fixed-effects instead of a Logit model because its coefficients are easier to interpret, especially in the presence of interaction terms where the derivation of marginal effects can be non-trivial (Ai and Norton, 2003; Greene, 2010). We do not consider the fixed-effects Probit model given its known bias (Greene, 2004).

Table 4: Myopic rationality of individual decisions

	Turns				Mean differences†
	[1-6]	[7-12]	[13-18]	≥ 19	$\Delta_{[7/12]-[13/18]}$
A. All	0.80	0.81	0.89	0.94	-0.083***
	(0.007)	(0.007)	(0.007)	(0.010)	
B. By decision problem					
i. Stay unlinked (Pass)	0.39	0.38	0.54	0.56	-0.156***
	(0.031)	(0.024)	(0.039)	(0.097)	
ii. Stay linked (Pass)	0.89	0.87	0.89	0.95	-0.019
	(0.009)	(0.015)	(0.015)	(0.017)	
iii. Remove link (Act)	0.75	0.83	0.91	0.95	-0.084***
	(0.011)	(0.009)	(0.009)	(0.012)	
iv. Propose link (Act)	0.98	0.99	1.00	0.98	-0.011
	(0.007)	(0.005)	(0.003)	(0.017)	
C. By treatment					
Treatment **N**	0.83	0.83	0.88	0.94	-0.058***
	(0.011)	(0.011)	(0.012)	(0.014)	
Treatment $\mathbf{S}^-$	0.81	0.80	0.89	0.97	-0.085***
	(0.012)	(0.012)	(0.012)	(0.018)	
Treatment $\mathbf{S}^+$	0.76	0.79	0.90	0.92	-0.107***
	(0.013)	(0.012)	(0.012)	(0.018)	

Std. errors of means in parenthesis. †Two-sided t-test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Movements to and from the PNS network

Turns	Enter	Stay	Leave	Total turns
Turns [1-12]				
Treatment $\mathbf{S}^-$	7	17	4	384
Treatment $\mathbf{S}^+$	30	69	24	384
Turns [$\geq$ 13]				
Treatment $\mathbf{S}^-$	8	19	1	128
Treatment $\mathbf{S}^+$	9	44	0	143

vector of attributes. Meanwhile, c_n captures the unobservable characteristics of subject n which may affect how she makes decisions. We do not assume that the unobservable individual characteristics are independent from the attributes of the decisions, and hence, implement an individual fixed effects specification. The standard errors are clustered by session. At the end of the section, we briefly discuss some extensions and alternative representations.

We can use the regression framework to investigate the four types of decisions described in Panel B of Table 4. Consider first the following simple specification:

$$\mathbb{E}(Y_{nt}^{ij} \mid \mathbf{X}) = \beta_0 + \beta_1 \cdot morelink_{ij} + \beta_2 \cdot act_{ij} + \beta_3 \cdot (morelink_{ij} \times act_{ij}) + \varepsilon \qquad (2)$$

where $morelink_{ij}$ and act_{ij} are dummy variables and ε is the residual. The variable $morelink_{ij}$ takes on a value of 1 if between networks i and j the network with more links gives the individual a higher payoff. The variable act_{ij} takes on a value of 1 if the myopic rational choice is to act.

Under the LPM, the interpretation of these β-coefficients is straightforward. The coefficient β_0 captures the probability that a subject stays unlinked in accordance to the myopic rational strategy (M. rat.). Similarly, $\beta_0 + \beta_1$ captures the probability that a subject stays linked in accordance to the myopic rational strategy. Table 6 provides interpretations for the different combinations of coefficients.

Table 6: The regression coefficients and the types of decision problems

	Interpretation	$more\ link_{ij}$	act_{ij}	Function
i.	$\mathbb{P}$(M. rat. | M. rat. action = stay unlinked)	0	0	β_0
ii.	$\mathbb{P}$(M. rat. | M. rat. action = stay linked)	1	0	$\beta_0 + \beta_1$
iii.	$\mathbb{P}$(M. rat. | M. rat. action = remove link)	0	1	$\beta_0 + \beta_2$
iv.	$\mathbb{P}$(M. rat. | M. rat. action = propose link)	1	1	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

We extend this basic specification with three sets of additional variables (and the individual fixed effects) to explore individual strategies. The specification for the extended model is:

$$\begin{aligned}\mathbb{E}(Y_{nt}^{ij} \mid \mathbf{X}) = \beta_0 + \beta_1 \cdot morelink_{ij} + \beta_2 \cdot act_{ij} + \beta_3 \cdot (morelink_{ij} \times act_{ij}) \\ + \gamma \cdot mpay_{ij} + \textstyle\sum_{t=1}^{4} \chi_t \cdot turn_sp(t) + c_n + \varepsilon\end{aligned} \qquad (3)$$

where $mpay_{ij}$ denotes the marginal payoff from making a myopic rational choice to evolve from network i to network j. We also include a linear spline on the turn variables, $turn_sp$,

with knots at turns 6, 12, and 18 to control for possible turn effects.[21] The knot choices mimic the turn grouping we did for the descriptive analysis.

5.2.2 Hypothesis and results

Hypothesis 2 *Subjects are more likely to follow the improving path:*

(*a*) *After Turn 12;*

(*b*) *When the myopic rational action increases the number of links;*

(*c*) *When the marginal loss from a deviation is larger.*

These hypotheses apply across treatments. Hypothesis 2(a) posits that behavior may change when the current turn can potentially be the final one. It can be tested by determining if there is a structural change after Turn 12. Hypothesis 2(b) suggests that the asymmetry in the (bilateral) formation and (unilateral) deletion of links may influence the strategy of subjects. It can be tested by determining whether the coefficients *morelink* and ($morelink \times act$) are positive and significant. Finally, Hypothesis 2(c) builds on the idea that marginal payoffs from deviations may affect decisions to stray from the improving paths. Indeed, with a myopic strategy, only the sign (a loss vs. a gain) but not the magnitude of the payoff should matter. However, if we assume imperfect choices (analogous to those assumed in random utility models or the Quantal Response Equilibrium model of McKelvey and Palfrey (1995) for example) it is reasonable to think that deviations from the improving path are less likely to occur when marginal losses are large. This can be tested by determining whether the coefficient $mpay_{ij}$ is positive and significant.

With these premises in mind, we next turn to test each hypothesis. The results can be summarized as follows.

Result 2 *Our analysis shows that:*

(a) Actions are more myopic rational after Turn 12.

(b) In early turns, subjects deviate from improving paths by maintaining excessive links (over-proposing and not removing redundant links). In later turns, subjects deviate by not removing redundant links.

(c) The size of marginal payoffs affects the likelihood of a deviation from myopic rationality in early turns of all treatments.

[21] Hence, the variable $turn_sp(1)$ is the spline for Turns [1-6], $turn_sp(2)$ is for Turns [7-12], $turn_sp(3)$ is for Turns [13-18] and $turn_sp(4)$ is for Turns 19 and greater.

We first performed a test to investigate whether there is a structural break at Turn 12 of each treatment. We implemented a standard test of pooling for models based on both Equations 2 and 3.[22] We cannot reject the null hypothesis that, on average, subjects altered their behaviors after Turn 12 (see Appendix B.3 for details). We therefore confirm differential behaviors before and after Turn 12, and hereafter, separately analyze decisions in Turns [1-12] and Turns [$\geq$ 13].

Our analysis begins by examining the extent to which improving paths drive individual behaviors. If improving paths were the sole driver of network evolution, the constant term in all specifications (β_0) would be one and the coefficients on all other variables ($\beta_1, \beta_2, \beta_3$) would be all zero. Table 7 presents the regression results of our basic model with individual fixed effects. The constant terms are high but significantly lower than one, and the coefficients of the other variables are significantly different from zero, suggesting deviations from the improving paths.

To better study deviations, we included estimates of the linear combinations of the coefficients for the *constant*, *morelink*, *act*, and *morelink* $\times$ *act*. These linear combinations are derived from Table 6 to allow immediate comparisons of the probabilities that individuals make myopic rational choices for the different decision problems. Pairwise comparisons of estimates confirm that, all else the same, subjects are more myopic rational in Turns [$\geq$ 13] than in Turns [1-12]. In all 12 combinations of treatments and decision problems, the point estimates are always larger in later turns, although the differences are not always statistically significant (in particular, when myopic rationality is close to 1 in early turns, the increase is necessarily limited). Overall, the regression provides support for the existence of a structural break in the proportion of myopic rational decisions after Turn 12 in all treatments, as conjectured in Hypothesis 2(a).

We next investigate myopic rationality as a function of the decision problem (pass / remove / propose). Panel B of Table 4 suggests that subjects keep too many links. In Hypothesis 2(b), we argued that the asymmetry of the linking game may explain this behavior. Since link formation requires mutual agreement while removal does not, one possible strategy would be to form and maintain some redundant links early on. As the game approaches the end, subjects begin to unilaterally remove some of them.

We find some evidence supporting this hypothesis in our regressions. As shown in Table 7, in Turns [1-12] the coefficient for myopic rationality in all treatments is highest

[22] Formally, we estimated a model where each regressor of interest was interacted with an indicator variable of whether an observation comes after Turn 12. A rejection of a joint test of the null hypothesis that all these interacted variables equal to zero is evidence for the presence of a structural break at Turn 12.

Table 7: FE LPM on myopic rationality: effect of type of decision

	Turns [1-12]			Turns [$\geq$ 13]		
	N	$\mathbf{S}^-$	$\mathbf{S}^+$	**N**	$\mathbf{S}^-$	$\mathbf{S}^+$
	(1)	(2)	(3)	(4)	(5)	(6)
morelink [β_1]	0.168***	0.144***	0.233***	0.040	0.072**	0.097**
	(0.026)	(0.022)	(0.010)	(0.031)	(0.029)	(0.033)
act [β_2]	-0.265***	-0.512***	-0.277***	-0.374***	-0.418***	-0.168***
	(0.062)	(0.027)	(0.047)	(0.105)	(0.077)	(0.046)
morelink $\times$ act [β_3]	0.173***	0.444***	0.263***	0.287*	0.335***	0.201***
	(0.045)	(0.025)	(0.055)	(0.130)	(0.062)	(0.050)
Constant [β_0]	0.809***	0.816***	0.731***	0.942***	0.921***	0.878***
	(0.017)	(0.012)	(0.006)	(0.020)	(0.018)	(0.010)
Linear combinations						
i. β_0	0.809***	0.816***	0.731***	0.942***	0.921***	0.878***
[stay unlinked]	(0.017)	(0.012)	(0.006)	(0.020)	(0.018)	(0.010)
ii. $\beta_0 + \beta_1$	0.977***	0.960***	0.964***	0.981***	0.993***	0.976***
[stay linked]	(0.020)	(0.012)	(0.008)	(0.014)	(0.027)	(0.030)
iii. $\beta_0 + \beta_2$	0.544***	0.303***	0.454***	0.568***	0.503***	0.711***
[remove link]	(0.047)	(0.019)	(0.044)	(0.094)	(0.072)	(0.044)
iv. $\beta_0 + \beta_1 + \beta_2 + \beta_3$	0.885***	0.892***	0.950***	0.894***	0.909***	1.009***
[propose link]	(0.015)	(0.015)	(0.019)	(0.030)	(0.046)	(0.039)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2304	2304	2304	972	768	858

Standard errors clustered at the session level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

when the myopic rational action is to stay linked (ii), followed by propose a link (iv), stay unlinked (i), and remove a link (iii). For Turns [$\geq$ 13], subjects are most likely to deviate by not removing a link when they should (iii). The order of the other coefficients are somewhat perturbed although it is difficult to make strong conclusions since most coefficients are very high (90 percent and above). Nonetheless, this evidence suggests that subjects use redundants links as a form of insurance, as was found in Deck and Johnson (2004).

We also study the effect of payoff magnitudes on choices. With a myopic rational strategy, the extent of the marginal payoff should be irrelevant: subjects would choose actions that give them non-zero gain, irrespective of their size. This contrasts with stochastic choice theories (random utility model, quantal response equilibrium, etc.) where mistakes are less prevalent if the associated costs are higher. To investigate this possibility in our setting, we implement the extended specification of (3). We then interact the payoff variable with the interactions between *morelink* and *act* to capture differential effects of marginal payoffs across different decision problems.

The results of the regressions are presented in Table 8. We linearly combine the coefficients for the payoff variables to explore the heterogeneity of the payoff-size effects across decision problems. We use a strategy similar to the way we linearly combined in Table 6 the coefficients of the *morelink*, *act* and (*morelink* $\times$ *act*) to examine the myopic rationality of the different decision problems. Hence, for example, γ_0 measures how the size of the marginal payoff affects the probability that subjects take the myopic rational choice to stay unlinked; $\gamma_0 + \gamma_1$ measures how the size of the marginal payoff influences the probability that subjects take the myopic rational choice to stay linked, and so on.

For Turns [1-12] in all treatments, myopic rational actions are positively and very significantly correlated with the size of marginal payoffs if the myopic rational action reduces the number of links (cases (i) and (iii)). Only for Treatment $\mathbf{S}^-$ it is also correlated if the myopic rational action increases the number of links. Two out of six of these coefficients lose their significance in Turns [$\geq$ 13]. These results support Hypothesis 2(c) and provide additional insights on how subjects deviate from improving paths. The evidence suggests that subjects pay more attention (and react more) to the opportunity loss from removing a link than from staying unlinked, especially earlier in the game.

In Appendix B.4, we also tested for a possible role of learning. As discussed above, subjects in each session play the same treatment twice. We examined differences in behavior between the first and second time a treatment was encountered by running a test of pooling, akin to the test of a structural break at Turn 12 described above. In general, we find little evidence of a change in behavior. The most notable difference is a slight increase in the

Table 8: FE LPM on myopic rationality: effect of marginal payoff

	Turns [1-12]			Turns [$\geq$ 13]		
	N	$\mathbf{S}^-$	$\mathbf{S}^+$	**N**	$\mathbf{S}^-$	$\mathbf{S}^+$
	(1)	(2)	(3)	(4)	(5)	(6)
morelink	0.335***	0.496***	0.355***	0.156	0.118**	0.151
	(0.053)	(0.141)	(0.044)	(0.092)	(0.049)	(0.092)
act	-0.313**	-0.617***	-0.385***	-0.081	0.085	-0.389**
	(0.104)	(0.154)	(0.059)	(0.159)	(0.311)	(0.145)
morelink $\times$ act	0.217*	0.583***	0.471***	-0.026	-0.241	0.527**
	(0.113)	(0.153)	(0.045)	(0.172)	(0.349)	(0.187)
mpay [γ_0]	0.012**	0.029**	0.018***	0.011**	0.006**	0.012*
	(0.004)	(0.010)	(0.003)	(0.004)	(0.002)	(0.005)
mpay $\times$ morelink [γ_1]	-0.013***	-0.026**	-0.017***	-0.010	-0.005	-0.010
	(0.004)	(0.010)	(0.003)	(0.006)	(0.004)	(0.006)
mpay $\times$ act [γ_2]	0.008	0.017	0.010	-0.032	-0.043	0.026
	(0.009)	(0.011)	(0.006)	(0.024)	(0.029)	(0.015)
mpay $\times$ morelink $\times$ act [γ_3]	-0.008	-0.017	-0.013*	0.033	0.047	-0.031*
	(0.009)	(0.012)	(0.006)	(0.023)	(0.031)	(0.015)
Constant	0.652***	0.390**	0.502***	0.750***	0.803***	0.775***
	(0.060)	(0.139)	(0.048)	(0.072)	(0.041)	(0.049)
Linear combinations						
i. γ_0	0.012**	0.029**	0.018***	0.011**	0.006**	0.012*
[stay unlinked]	(0.004)	(0.010)	(0.003)	(0.004)	(0.002)	(0.005)
ii. $\gamma_0 + \gamma_1$	-0.001	0.003**	0.002	0.001	0.001	0.001
[stay linked]	(0.001)	(0.001)	(0.001)	(0.002)	(0.003)	(0.003)
iii. $\gamma_0 + \gamma_2$	0.020**	0.046***	0.028***	-0.021	-0.037	0.038**
[remove link]	(0.006)	(0.007)	(0.006)	(0.025)	(0.028)	(0.012)
iv. $\gamma_0 + \gamma_1 + \gamma_2 + \gamma_3$	-0.000	0.002**	-0.002	0.002	0.005	-0.004
[propose link]	(0.003)	(0.001)	(0.002)	(0.002)	(0.005)	(0.005)
Turn spline variables	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2304	2304	2304	972	768	858

Standard errors clustered at the session level in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

response to marginal losses the second time subjects played Treatment **N**.

5.3 Model predictions

We next examine the capacity of our empirical model to *predict* actions. Our prediction model uses a relatively parsimonious specification that incorporates insights from Result 2. In particular, we estimated a logit model where we supplanted the basic model of Equation 1 with (i) $mpay$ and (ii) $(mpay \times morelink)$ to incorporate Results 2(b) and 2(c). To incorporate the insight from Result 2(a) and account for the structural break at Turn 12, all of these variables are interacted with the indicator variable $\mathbb{1}(turn > 12)$. To maintain model parsimony, we do not include the individual fixed effects.

We use the model to conduct an out-of sample prediction exercise. For each treatment, we estimated the coefficients of the model with a sample that excludes observations from that treatment. Once the coefficients are recovered, we use the model to predict the actions in the excluded treatment. Table 9 shows the coefficients from the estimation exercise. We then use these coefficients to predict the out-of-sample actions of the first 18 turns. Figure 2 graphically depicts for each treatment the plot of the myopic rationality of actual choices (dashed line), out-of-sample predicted choices (solid line), and 95 percent confidence interval of predicted choices.

The model generally predicts actions well, even though it performs less well in predicting initial behavior and switches around Turn 12. Indeed, in all three treatments actions are more likely to be outside the prediction intervals in the first few turns and the turns around Turn 12. However, the absolute differences are always relatively small.

5.4 Summary

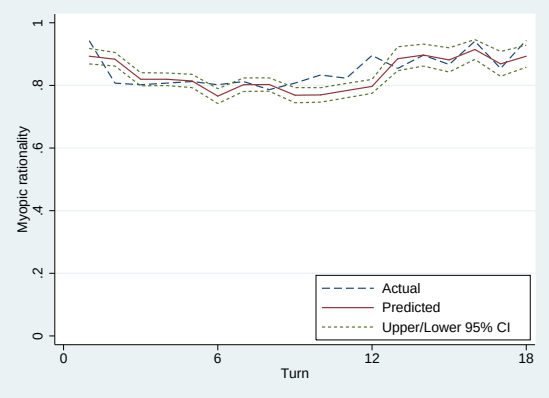
The analysis at the single decision level suggests that subjects take for the most part the myopic rational action. At the same time, we highlight important and *systematic* deviations. Indeed, we observe less myopically-rational actions in turns with a sure future than in random-ending turns. Deviations also tend to take the form of excessive links, possibly because they can be removed unilaterally, although proving this hypothesis would require further work. Finally, deviations are also more prevalent the smaller the marginal payoff losses, as expected in a behavioral theory where "mistakes" (which could be part of experimentation) depend inversely on loss magnitudes. Overall and with some interesting exceptions, the analysis provides support for convergence to PNS networks through myopic rational choices.

Table 9: Logit regression

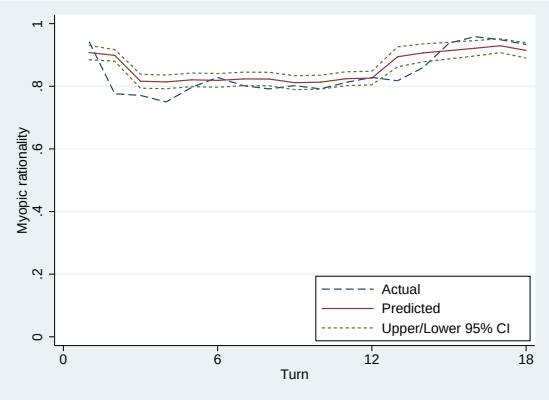
	N	**S**$^-$	**S**$^+$
	(1)	(2)	(3)
$\mathbb{1}$(turn>12)	1.984***	1.594***	1.908***
	(0.239)	(0.219)	(0.370)
morelink	3.575***	4.687***	3.091***
	(0.332)	(0.485)	(0.331)
... × $\mathbb{1}$(turn>12)	-1.449*	-1.322	-1.185
	(0.855)	(1.153)	(0.760)
act	-2.043***	-1.449***	-1.720***
	(0.118)	(0.111)	(0.115)
... × $\mathbb{1}$(turn>12)	-0.128	-0.658***	-0.853***
	(0.252)	(0.238)	(0.254)
morelink × act	0.263	-1.326***	0.057
	(0.321)	(0.473)	(0.291)
... × $\mathbb{1}$(turn>12)	-0.635	0.160	0.035
	(0.846)	(1.141)	(0.706)
mpay	0.094***	0.080***	0.082***
	(0.009)	(0.009)	(0.013)
... × $\mathbb{1}$(turn>12)	-0.088***	-0.036*	-0.049*
	(0.023)	(0.022)	(0.029)
mpay × morelink	-0.082***	-0.094***	-0.070***
	(0.014)	(0.013)	(0.016)
... × $\mathbb{1}$(turn>12)	0.120***	0.066**	0.059*
	(0.034)	(0.030)	(0.034)
Constant	0.271***	0.431***	0.441**
	(0.103)	(0.097)	(0.181)
Observations	6234	6438	6348
Pseudo R^2	0.197	0.162	0.161

Standard errors clustered at the session level in parentheses.

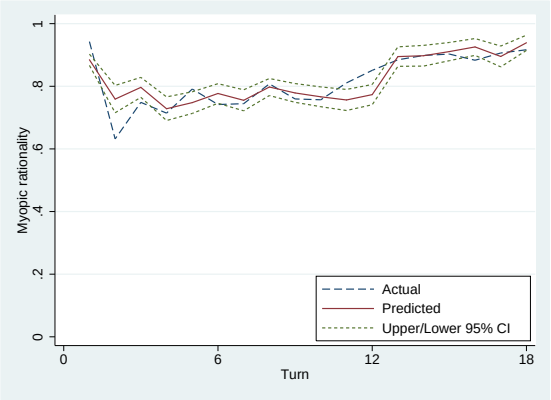
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$



(a) Treatment **N**



(b) Treatment $\mathbf{S}^{-}$



(c) Treatment $\mathbf{S}^{+}$

Figure 2: Out-of-sample logit prediction by treatment

6 Choices by subjects

So far we have analyzed network outcomes and single decisions. One question that remains unaddressed is the degree of *subject* heterogeneity. To answer that question, we use cluster analysis to classify subjects based on their behavior. We implemented the mixture models approach, which treats each cluster as a component probability distribution and endogenously chooses (optimally) the model and number of clusters using Bayesian statistical methods (Fraley and Raftery, 2002). In doing so, it avoids the necessity of having to set the number of clusters and clustering criterion *ex ante* that was found in alternative clustering approaches. We implement model-based clustering analysis with the *mclust* package in R (Fraley et al., 2012). A maximum of nine clusters are considered for up to ten different models and the combination yielding the maximum Bayesian Information Criterion (BIC) is chosen.

Since we want to study heterogeneity across subjects in the likelihood of choosing myopic rational strategies, the variables we use as inputs to the model are guided by the findings in Result 2: subjects' level of myopic rationality at early and late turns, and subjects' willingness to deviate given the size of the marginal loss, also at early and late turns. Indeed, our analysis of individual decisions suggests statistically significant jumps in myopic rationality between early and late turns. To capture subject level variation, we consider myopic rationality in early turns (1 to 12) and late turns (13 and above) separately. Meanwhile, to capture the responsiveness of each subject's deviation to the size of marginal loss, we estimated a regression akin to Equation (3) for each of the subjects, and use the subject's coefficient on the marginal payoff (γ) as this measure of responsiveness to the marginal loss.[23] The coefficients are estimated for early and late turns.

Hypothesis 3 *There are three types of subjects: (i) random (with low levels of myopic rationality), (ii) rational (with high levels of myopic rationality) and (iii) strategic (who deviate from myopic rationality only when the cost is low). Earnings are lowest for random subjects and highest for strategic subjects.*

We anticipate substantial heterogeneity across individuals. In a game where option values are difficult to compute, trading-off current costs and benefits (as myopic rationality predicts) seems a plausible, reasonably sophisticated strategy. Since the game is inherently difficult, we also expect to observe some subjects to be "lost in the network." Finally, the most interesting behavior relates to subjects who realize the appeal of myopic rational

[23] The equation estimated is slightly different from (3) in that we dropped the turn splines variables and we allowed the coefficients on the regressors to vary between early and late turns.

choices but also try to exploit its shortcomings. These subjects will deviate early in the game and when the marginal loss is low. They are also expected to accumulate the highest earnings.

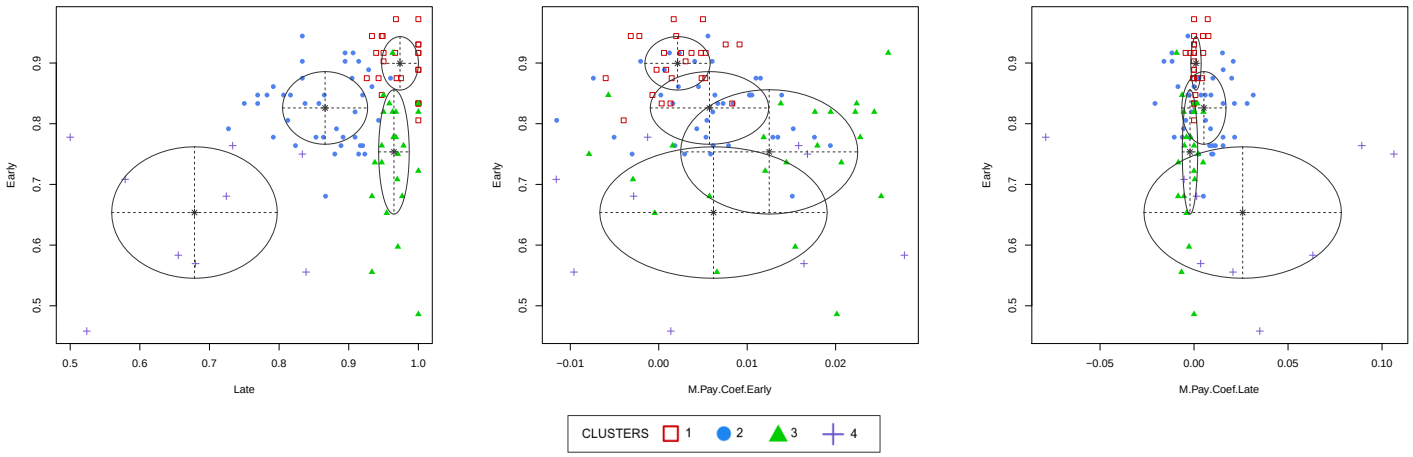
Result 3 *We observe 10 percent of random subjects, 25 percent of rational subjects and 65 percent of subjects with two different levels of strategic behavior. Earnings are monotonically increasing in the proportion of myopic rational choices in early turns.*

The model endogenously generated *four* clusters. Figure 3 shows the model's projections onto three variable pairs. On the vertical axis, we present the most discriminatory variable, namely myopic rationality in early turns. On the horizontal axis, we present each of the other three variables: myopic rationality in late turns (3a); marginal payoff coefficient in early turns (3b); and marginal payoff coefficient in late turns (3c). Table 10 provides a summary of the mean values of the input variables and earnings within each cluster, sorted by their mean earnings.

As we can see from Figure 3a and Table 10 (Columns 2-3), the four clusters are clearly differentiated in terms of the level of myopic rationality in early and late turns. About 25 percent of subjects (Cluster 1) are "rational": they consistently followed the improving path throughout the game. On the other end, 10 percent of the subjects (Cluster 4) are "random": they often deviate from the improving path, playing the myopic strategy barely more frequently than predicted by chance. The majority of subjects, around 65 percent (Clusters 2 and 3), are "strategic": they explore strategies involving non-myopic rational choices in early turns before resorting to primarily myopic rational choices in late turns.

Figure 3b and Table 10 (Column 4) suggest that these subjects can be further classified based on the marginal payoff coefficients in early turns. Indeed, subjects in Cluster 3 are more likely to use the magnitude of the marginal losses to guide their strategy in early turns before entirely abandoning it in late turns. In comparison, Cluster 2 subjects rely less on the marginal payoff coefficient: Their mean level of myopic rationality (marginal payoff coefficients) are higher (lower) than Cluster 3 subjects' in early turns, but are stable throughout the game. Finally, notice from Figure 3c and Table 10 (Column 5) that unlike the others, subjects in Cluster 4 fail to realize the urgency of making myopic rational choices following Turn 12. Against the expectation of profit maximizing subjects, they respond more to marginal losses in later than in earlier turns of the game. This is further evidence that these subjects do not fully grasp the trade-offs of the game.

Our decomposition exercise confirms the distinctions between clusters whose subjects consistently followed the improving path, consistently ignored the improving path, and



(a) (b) (c)

Figure 3: BIC-Maximizing Clusters of the Mixture Models

Table 10: Subject-level mean behaviors by cluster

Cluster	Myopic rational		M. Payoff Coeff.		Earnings	N
	[1-12]	[≥ 13]	[1-12]	[≥ 13]		
1	0.90	0.97	0.002	0.001	83.6	24
	(0.009)	(0.006)	(0.001)	(0.001)	(4.05)	
2	0.83	0.86	0.006	0.005	76.1	40
	(0.009)	(0.009)	(0.001)	(0.002)	(3.49)	
3	0.74	0.97	0.013	-0.002	72.3	23
	(0.021)	(0.004)	(0.002)	(0.001)	(4.89)	
4	0.65	0.67	0.006	0.026	71.7	9
	(0.037)	(0.041)	(0.005)	(0.019)	(7.38)	

Columns 2-5 are *subject-level* means. Standard errors of means in parenthesis.

the majority who explored some (early) deviations. In Table 11, we study whether these differences across clusters are similar in all three treatments. The patterns are remarkably similar in **N**, $\mathbf{S}^-$, and $\mathbf{S}^+$. Perhaps the only noticeable difference is that in Treatment $\mathbf{S}^-$ Cluster 2 subjects exhibit a significant increase in myopic rationality between early and late turns, much like Cluster 3 subjects.

Table 11: Myopic rationality of decisions by cluster and treatment

Cluster	1		2		3		4	
Turn	[1-12]	[13-18]	[1-12]	[13-18]	[1-12]	[13-18]	[1-12]	[13-18]
Treatment								
Treatment **N**	0.88	0.98	0.83	0.86	0.82	0.92	0.71	0.63
	(0.013)	(0.010)	(0.012)	(0.020)	(0.016)	(0.020)	(0.031)	(0.062)
Treatment $\mathbf{S}^-$	0.90	0.93	0.82	0.84	0.74	0.96	0.68	0.81
	(0.013)	(0.020)	(0.012)	(0.022)	(0.019)	(0.014)	(0.032)	(0.051)
Treatment $\mathbf{S}^+$	0.91	0.99	0.83	0.87	0.67	0.97	0.56	0.62
	(0.012)	(0.006)	(0.012)	(0.022)	(0.020)	(0.013)	(0.034)	(0.062)
All	0.90	0.97	0.83	0.86	0.74	0.95	0.65	0.68
	(0.007)	(0.008)	(0.007)	(0.012)	(0.011)	(0.010)	(0.019)	(0.034)

Numbers are *decision-level* means. Standard errors of means in parenthesis.

From Table 10 we also notice that earnings are positively correlated with the proportion of myopic rational choices in early turns. This contrasts with our hypothesis that the more strategic subjects – those who do not necessarily take myopic rational decision early in the game but do it consistently later on – would obtain the highest profits.[24]

A possible explanation is that payoffs in the game are (very) noisy signals of the strategy of individuals: It depends on the behavior of the five other subjects in the network and their final position in the component. The most forward looking subjects may end up bearing a larger number of links and/or being negatively affected by subjects who use suboptimal strategies. To investigate the effect of subject composition on network outcomes, we count for each match the number of subjects from each cluster. We then regress whether the match ended in a cycle (for Treatment **N**) or in the PNS network (for Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$) on the number of subjects from each cluster. Cluster 1, whose subjects' behavior is most similar to those assumed in theory, are used as the benchmark and is therefore the omitted category in the regression. The models are estimated with a session fixed effects and results are presented in Table 12.

[24]The result, however, should not be overemphasized since pairwise t-tests do not find statistically significant differences in mean earnings between clusters except for Cluster 1.

Table 12: Cluster and network outcomes

	$\mathbf{N}$	$\mathbf{S}^-$	$\mathbf{S}^+$
	(1)	(2)	(3)
Cluster 2	-0.26	-0.08	-0.08
	(0.14)	(0.15)	(0.10)
Cluster 3	-0.05	0.17	-0.10*
	(0.12)	(0.13)	(0.05)
Cluster 4	-0.47**	-0.04	-0.40***
	(0.14)	(0.16)	(0.10)
Constant	1.63**	0.29	1.05**
	(0.48)	(0.57)	(0.33)
Session FE	Yes	Yes	Yes
Observations	32	32	32
R^2	0.28	0.33	0.44

Standard errors clustered at the session level in parenthesis. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We find that the subject composition significantly affects the network outcome in Treatments $\mathbf{N}$ and $\mathbf{S}^+$, where the presence of Cluster 4 subjects (and to a lesser extent Cluster 3 in $\mathbf{S}^+$) negatively affects the likelihood of ending in the close cycle or the PNS network. By contrast, the proportion of different subject types does not influence the outcome in Treatment $\mathbf{S}^-$.

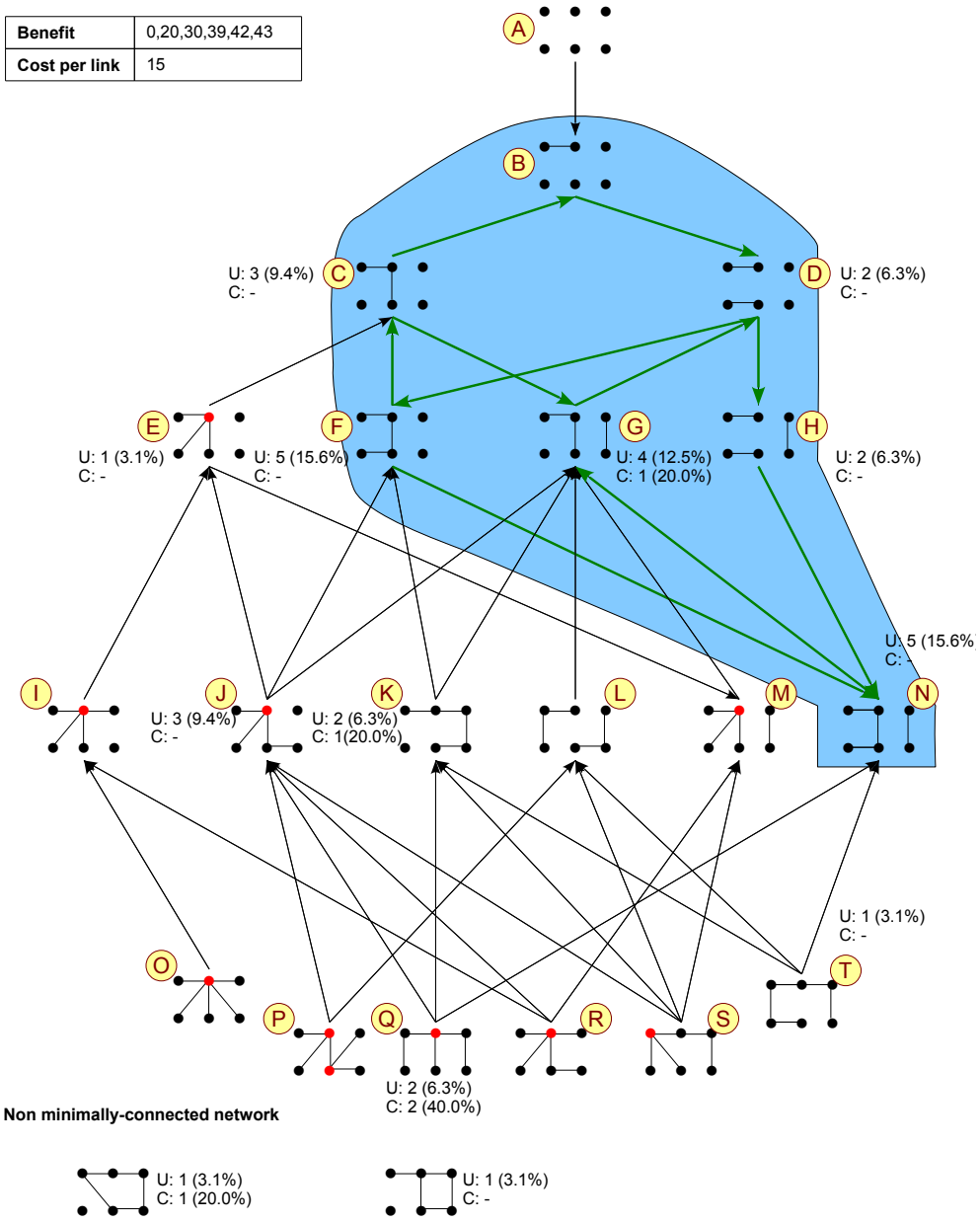
Overall, the cluster analysis highlights the significant heterogeneity in individual behavior, especially with respect to the level of myopic rational decisions. It also demonstrates the importance of group composition for network outcomes and earnings, with the presence of a random player significantly decreasing the likelihood of reaching the stable network or the close cycle.

7 Conclusion

The paper has studied the dynamic formation of social networks. Our subjects rarely consider the total value of the network as a key criterion when making their decisions. Instead, choices are largely consistent with individual maximization of payoffs, and the process typically converges to the PNS network if it exists. At the same time, not all PNS networks are equal, and a stronger strong stability notion predicts network outcomes significantly better. As for single decisions, although myopic rationality is predominant,

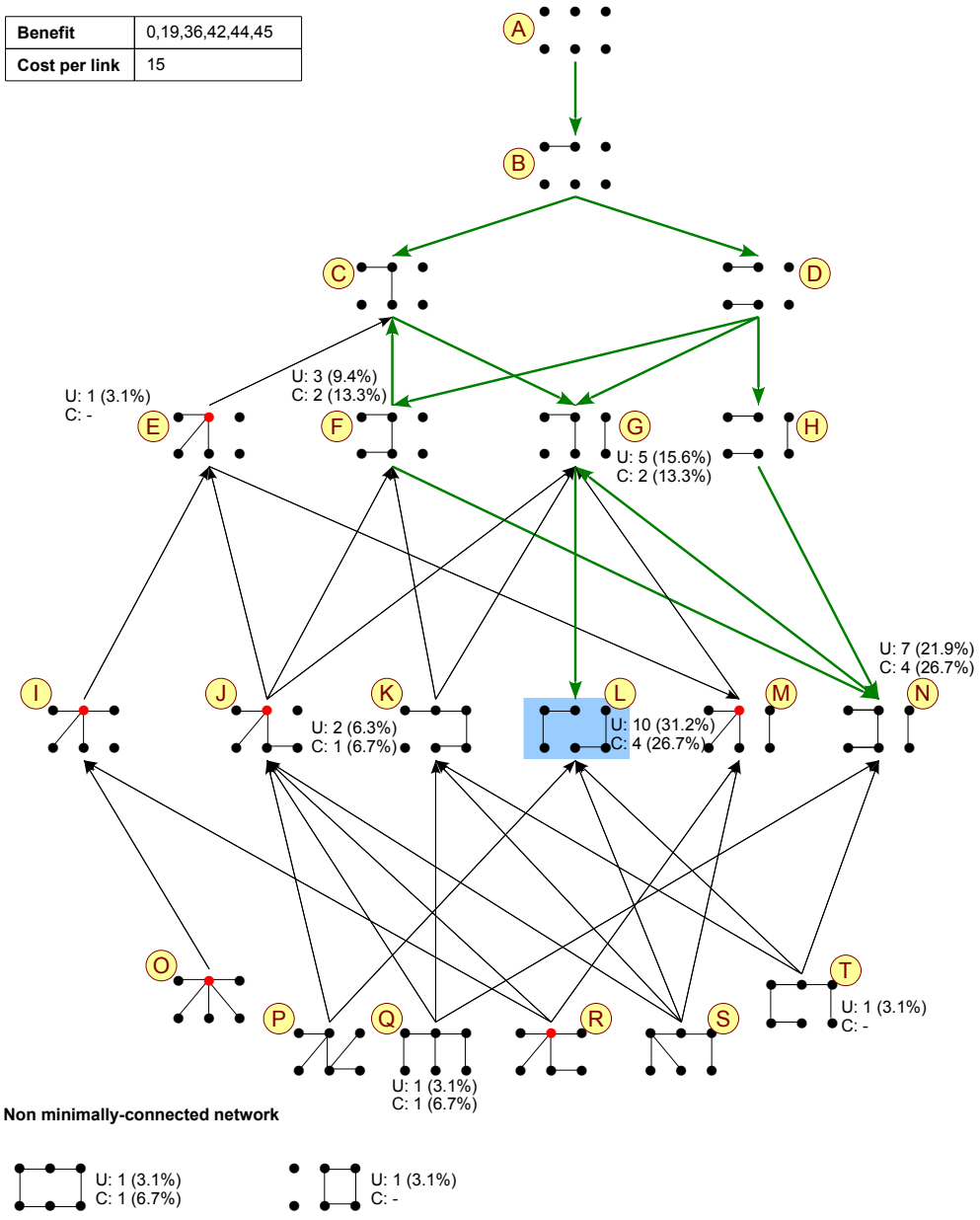
we also observe interesting systematic deviations from it. In particular, myopic rationality is less prevalent at the margin when actions are reversible, when marginal payoff losses are smaller and when actions involve excessive links that can be removed unilaterally later on. Finally, we also notice a significant heterogeneity in behavior across subjects, with a majority of subjects deviating from myopic rationality only early in the game.

Despite the recent advances, there is still much to learn about network formation, both theoretically and experimentally. On the theory front, it would be interesting to incorporate behavioral imperfections into existing models. The tendency observed in our data towards fewer deviations from myopic rationality as marginal losses increase and as matches enter the probabilistic ending phase suggests that subjects optimize subject to imperfect choice, imperfect foresight and/or imperfect understanding of the game. To our knowledge, however, no model has yet been developed to capture these frictions. On the experimental front, ecological validity is a concern. Indeed, we feel that our cost and benefit representation of adding and removing links captures the essence of social networks in an excessively stylized and abstract way. The use of laboratory studies in the field or laboratory studies that exploit social technologies (facebook, twitter, second life, etc.) would add a more realistic dimension to the network formation problem without compromising the controlled environment of the laboratory.



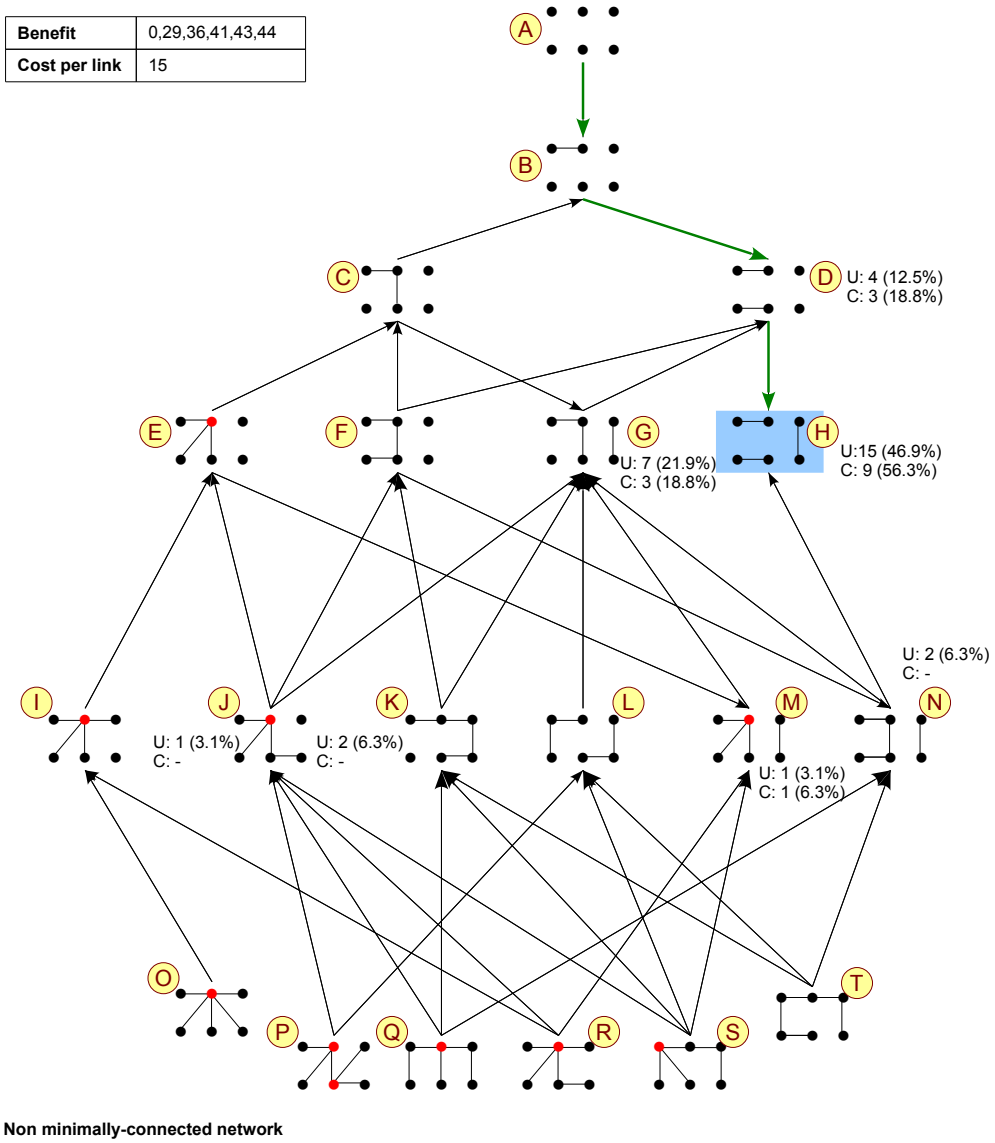
Notes. Letters refer to all minimally-connected network structures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. Networks that are part of the closed cycle are inside the shaded region.

Figure 4: No Pairwise-Stable Network and a Closed Cycle (Treatment **N**)



Notes. Letters refer to all minimally-connected network structures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. PNS network is shaded.

Figure 5: A Unique PNS (SSDM) Network (Treatment $\mathbf{S}^{-}$)



Notes. *Letters refer to all minimally-connected network structures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. PNS network is shaded.*

Figure 6: A Unique PNS (SSJN) Network (Treatment $\mathbf{S}^{+}$)

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APPENDIX (for online publication)

A Treatment M: Multiplicity of PNS networks

With Treatment **M**, we examine a game with multiple stable networks. We use the multiplicity of stable networks to examine whether subjects play some form of forward-looking behavior. A feature of the treatment is that subjects need to deviate from myopic rationality – and take the risk of incurring negative payoffs – in order to reach the (Pareto-dominant) stable network with strictly positive payoffs. This situation mimics the problem of network formation in scale economies, where the first few links have negative net benefits to the first adopters.

Figure A.1 depicts the payoff structure, improving paths, and stable networks in Treatment **M**. The treatment has four pairwise stable networks. Two of these are PNS networks $\{A, K\}$, where $\{A\}$ is stochastically stable while $\{K\}$ is SSDM. The remaining two ($\{I, J\}$) are not Nash stable. The payoffs of the SSDM network $\{K\}$ Pareto-dominate those in the empty PNS network $\{A\}$. The analysis is relegated to the appendix because our original intention was to build a network formation game with exactly two pairwise stable networks. Due to a mistake in programming, the game has four instead, which interferes with the analysis of the data and the interpretation of results. For transparency but also because the basic results are interesting, we summarize below the main findings.

The setup provides a direct test of the assumption of myopic rationality in network formation. To illustrate, the movement from network $\{A\}$ to $\{B\}$ requires that linking subjects be willing to risk receiving negative payoffs. If subjects primarily make myopic rational choices, then the network will be "stable" at A. However, if enough subjects are willing to deviate from the myopic rational strategy, they may reach the Pareto-dominant SSDM network $\{K\}$.

The set of deviations required to reach network $\{K\}$ from $\{A\}$ is non-trivial. Define the *resistance* between networks g and g' as the minimum number of mutations (i.e., deviations from an improving path) necessary to evolve from g to g' (Jackson and Watts, 2002). In Treatment **M**, the resistance between the starting network and the SSDM network equals to four.

We find evidence of strategic (forward-looking) behavior among the experimental subjects. At the network level, despite the non-trivial resistance, the dynamic link formation process did lead by and large to the SSDM network. Indeed, 44 percent (unconditional on convergence) and 55 percent (conditional on convergence) of matches in Treatment **M** ended in the SSDM network. By contrast, no match ended in the empty, stochastically-

stable PNS network. Meanwhile, only 9 percent (unconditional) and 5 percent (conditional) ended in one of the other two pairwise stable networks.

At the decision level, there is evidence of differences in myopic rationality across turns. Table A.1 presents the analogue of Table 4 for Treatment **M**. Interestingly, among the four treatments considered, only in Treatment **M** do we find a statistically significant increase in myopic rationality between Turns [1-6] and [7-12]. The low aggregate level of myopic rationality in the first few turns (0.56) is expected since such deviation is needed to "escape" from $\{A\}$, the low paying PNS network. At the same time, just like in the remaining three treatments, we also find a statistically significant increase in myopic rationality between Turns [7-12] and [13-18].

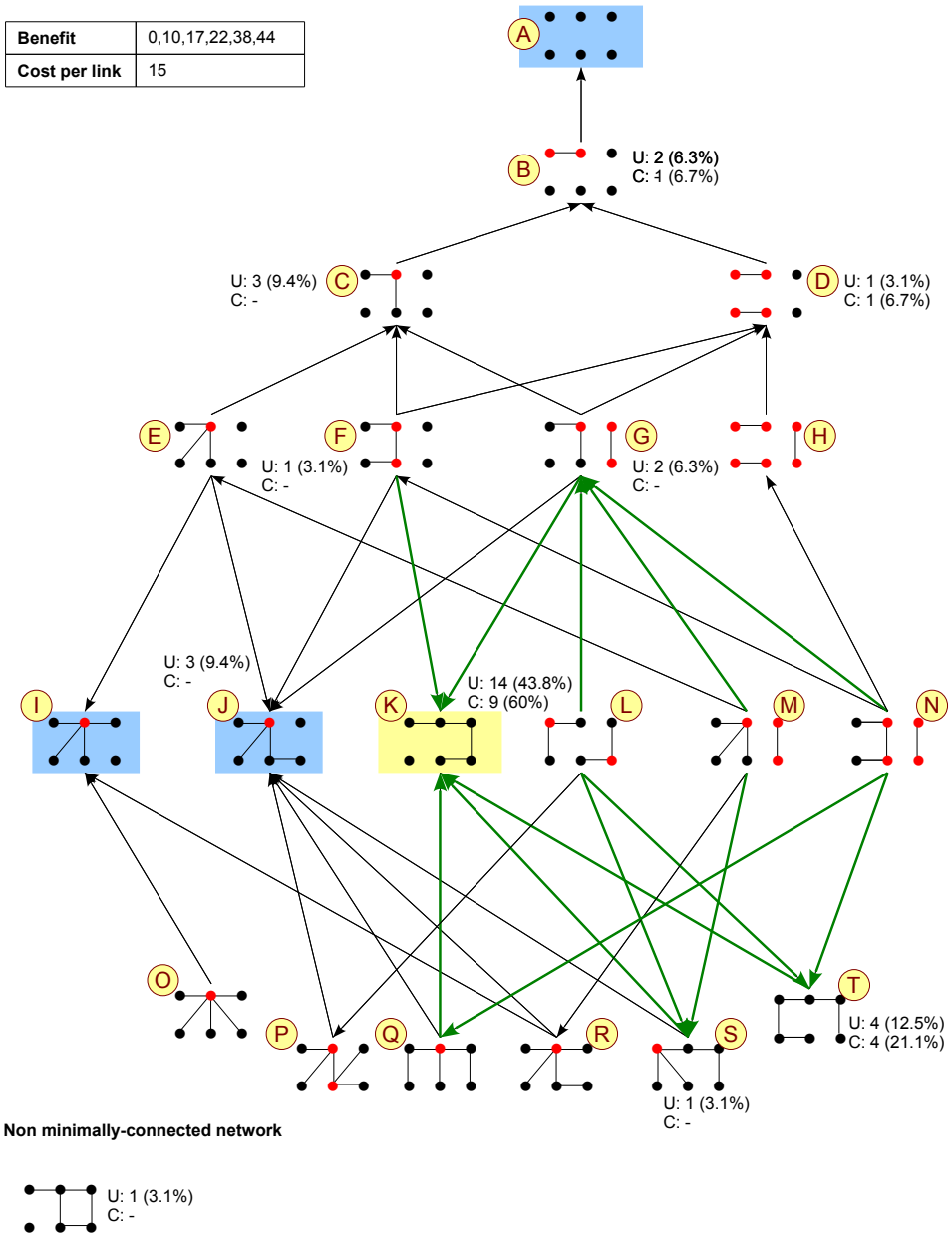
Table A.1: Myopic rationality of individual decisions

	Turns				Mean differences†	
	[1-6]	[7-12]	[13-18]	≥ 19	$\Delta_{[1/6]-[7/12]}$	$\Delta_{[7/12]-[13/18]}$
Treatment **M**	0.56	0.86	0.90	0.86	-0.293***	-0.049***
	(0.015)	(0.010)	(0.011)	(0.019)		

Std. errors of means in parenthesis. †Two-sided t-test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Our findings agree with Pantz (2006) who found subjects reaching the forward-looking network structure, and Kirchsteiger et al. (2013) who found evidence for limited farsightedness. However, compared to these other studies, our setup with six subjects is more complex for subjects. Subjects had a vast array of choices – including 20 minimally-connected structures – and had to overcome a significant resistance (4 mutations) to evolve the network from the starting stable network to the SSDM network. Moreover, they face a significant amount of uncertainty from random pairing and random ending. Even in this complex environment, agents systematically deviated from the improving paths, and are able to reach the SSDM network that would have been impossible to reach by primarily myopic-rational agents.

These results are also consistent with those described in the main body of the paper. Indeed, a substantial fraction of subjects take for the most part myopic rational actions, but they also *strategically* deviate under the appropriate circumstances: when the action is reversible and the marginal cost low (Treatments **N**, $\mathbf{S}^{-}$ and $\mathbf{S}^{+}$) and when they are required to escape a stable but low payoff network (Treatment **M**).



Notes. Letters refer to all minimally-connected network structures. Numbers next to each network refer to the frequency (and percentage) that the process ends in that network: U = Unconditional on convergence, C = Conditional on no change in the last 3 turns. Non-Nash PS networks are shaded pink, Non-SS PNS network is shaded blue, SSDM network is shaded yellow.

Figure A.1: Multiple Pairwise stable networks (Treatment **M**)

B Additional analyses

B.1 Geodesic distance to efficient and PNS networks

In this section, we provide a complementary study of the difference between observed and predicted outcomes to the one presented in section 4.1. More precisely, we calculate the shortest (or "geodesic") distance between the resulting networks and the closest network in the closed cycle (for Treatment **N**) or the PNS networks (for Treatments $\mathbf{S}^-$ and $\mathbf{S}^+$) as well as the distance between the resulting networks and the efficient networks. For the latter, we separately calculate the distance to the closest of all the efficient networks $\{O, P, Q, R, S, T\}$ and to the line network $\{T\}$.[1]

Panel A of Table B.1 shows that for Treatments **N** and $\mathbf{S}^+$, the distance to the closed cycle and the PNS network respectively is substantially shorter than the distance to the efficient networks. For Treatment $\mathbf{S}^-$, however, the distance from the PNS network is equal to the distance from the efficient line network and marginally longer than the distance from the closest efficient network, suggesting a larger dispersion in behavior.

Table B.1: Average distances from outcomes

Treatment	PNS	Closed cycle	Shortest Efficient	Efficient
	Panel A. **All**			
N	—	0.41 †	1.66	2.03
$\mathbf{S}^-$	1.41	—	1.28	1.41
$\mathbf{S}^+$	1.00	—	1.94	2.06
	Panel B. **Conditional on Convergence**			
$\mathbf{S}^-$	1.61	—	1.28	1.39
$\mathbf{S}^+$	0.88	—	2.06	2.18

Convergence defined as having no change in the final 3 turns.

†Distance to the closest network in the cycle.

Panel B of Table B.1 presents the average distance between outcomes and predicted networks conditional on convergence. For Treatment $\mathbf{S}^+$, the distance to the PNS network is even lower than unconditional on convergence, which strongly suggests the stability of

[1]If agents were to aim at the efficient network, the line network is the most likely outcome since it distributes payoffs most equally. For example $\{O\}$, which is never played in our experiment, is efficient but requires one player to form 5 links and therefore bear significant payoff losses (30 to 32 tokens depending on the treatment).

the SSJN network. In contrast, the distance from the convergent network to the efficient network is lower than to the PNS network in Treatment $\mathbf{S}^-$. Results from Panels A and B come from the fact that most of the network outcomes in Treatment $\mathbf{S}^-$, both conditional and unconditional on convergence, are split almost equally between the PNS network $\{L\}$ and network $\{N\}$. Since the distance between these two networks is two and both of them are at a distance of one to the efficient networks, the distance from the stable network and the efficient networks are similar. It is difficult to infer from the outcomes alone where the formation processes is leading toward. Our analysis of individual decisions (Result 2) provides a plausible explanation for these findings.

B.2 Myopic rationality by treatment and decision problem

Figure B.1 further illustrates the results in Panels B and C of Table 4. We plot for each treatment the proportion of myopic rational behavior across turns when passing is rational (B1) and when acting is rational (B2). In all treatments, players maintain more links in all turns than would be observed if they played myopically rational all the time. The gap is bigger and the variation larger for decisions where acting is myopic rational (figures on the right), although the difference narrows as the match nears its end.

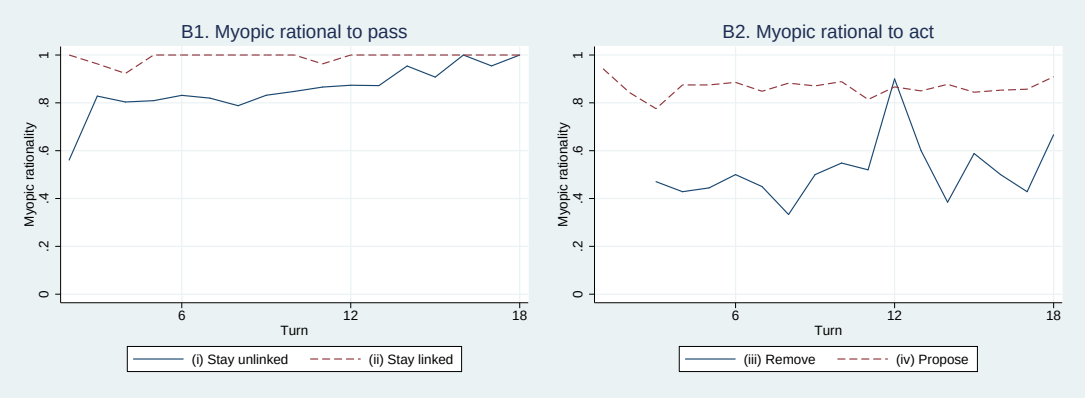
B.3 Pooled FE LPM on probability of myopic rational choices

To test whether there is a structural break in behavior at Turn 12, we implemented a test of pooling that is estimated using the fixed-effects LPM. For each treatment, we estimated models based on Equations 2 and 3, but interacted each of the variables with an indicator variable of whether an observation comes after Turn 12, $\mathbb{1}(turn > 12)$.

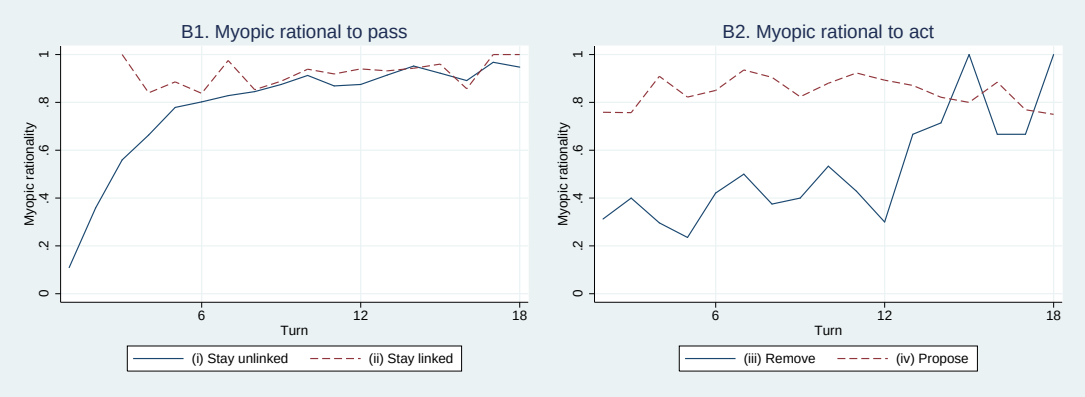
Table B.2 presents the fixed effects pooled LPM to investigate if there is a difference in myopic rational choices before and after Turn 12. To test whether there are structural breaks in the variables, we conducted a joint test of all interacted variables. The p-value of this test is presented at the bottom of the table. For all treatments and specifications, the null hypothesis of no structural break at Turn 12 is rejected at the 5 percent significance level.

B.4 Effect of experience

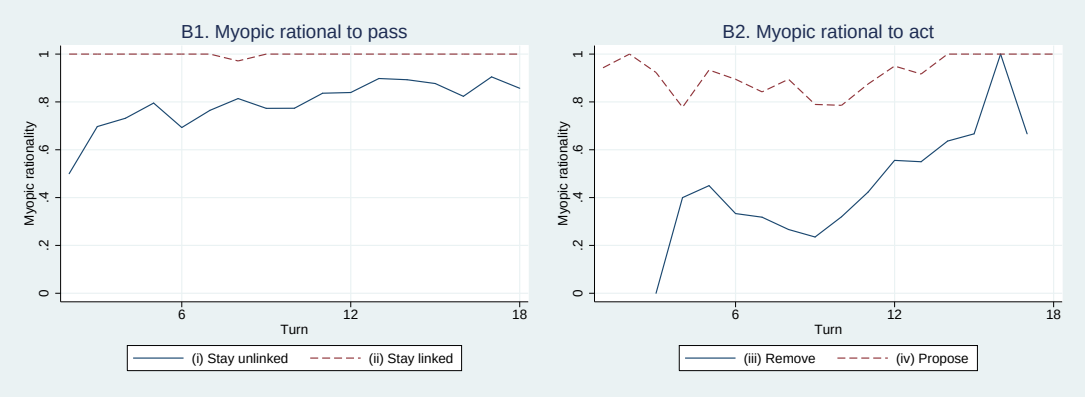
As described in Section 3, each subject played each treatment twice. We use this to examine whether subjects altered their behavior the second time they encountered a treatment. We implemented a test of structural break akin to the pooling test described in Section B.3.



(a) Treatment **N**



(b) Treatment $\mathbf{S}^{-}$



(c) Treatment $\mathbf{S}^{+}$

Figure B.1: Myopic rationality by treatment and decision problem

Table B.2: Pooled FE LPM on the likelihood of myopic rational action

	N		**S**$^-$		**S**$^+$	
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}(turn > 12)$	0.130***	0.056	0.105***	-0.008	0.169***	0.226**
	(0.023)	(0.044)	(0.017)	(0.055)	(0.042)	(0.076)
morelink	0.163***	0.165***	0.145***	0.137***	0.236***	0.180***
	(0.023)	(0.023)	(0.021)	(0.020)	(0.011)	(0.017)
$\ldots \times \mathbb{1}(turn > 12)$	-0.121***	-0.127***	-0.088***	-0.087***	-0.162***	-0.138**
	(0.029)	(0.030)	(0.019)	(0.019)	(0.038)	(0.045)
act	-0.274***	-0.269***	-0.524***	-0.513***	-0.294***	-0.303***
	(0.062)	(0.060)	(0.029)	(0.029)	(0.045)	(0.043)
$\ldots \times \mathbb{1}(turn > 12)$	-0.087	-0.081	0.103	0.107	0.077	0.091
	(0.082)	(0.088)	(0.063)	(0.064)	(0.061)	(0.058)
morelink × act	0.185***	0.197***	0.458***	0.477***	0.265***	0.315***
	(0.048)	(0.043)	(0.026)	(0.024)	(0.054)	(0.062)
$\ldots \times \mathbb{1}(turn > 12)$	0.079	0.051	-0.130**	-0.161**	-0.046	-0.098
	(0.092)	(0.106)	(0.052)	(0.047)	(0.066)	(0.069)
mpay		0.003**		0.005***		0.007***
		(0.001)		(0.001)		(0.002)
$\ldots \times \mathbb{1}(turn > 12)$		-0.000		-0.003		-0.003
		(0.003)		(0.002)		(0.002)
turn		0.004		0.003		0.010**
		(0.004)		(0.002)		(0.003)
$\ldots \times \mathbb{1}(turn > 12)$		0.003		0.009**		-0.007
		(0.004)		(0.003)		(0.005)
Constant	0.811***	0.748***	0.818***	0.727***	0.732***	0.602***
	(0.016)	(0.036)	(0.014)	(0.030)	(0.012)	(0.034)
Individual Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
$\mathbb{P}$(pooling)	0.004	0.014	0.003	0.005	0.023	0.011
Observations	3276	3276	3072	3072	3162	3162
Adj. R^2	0.169	0.174	0.253	0.260	0.245	0.260

Standard errors clustered at the session level in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We estimated models based on Equations 2 and 3, where the regressors are interacted with an indicator variable of whether an observation comes from the second half of the session, $\mathbb{1}$(second half).

Table B.3 presents the results regarding the effect of experience on behavior. In all but one specification (Treatment $\mathbf{S}^-$, extended model), we cannot reject the null hypothesis of no effect of experience (at the 5 percent level). Moreover, in terms of individual coefficients, only the *turn* coefficient for that specification is significantly different. Overall, there is very limited evidence of change in behavior between the first and second time the same game is played.

Table B.3: Pooled FE LPM on the likelihood of myopic rational action

	N		**S**$^-$		**S**$^+$	
	(1)	(2)	(3)	(4)	(5)	(6)
sechalf	-0.051	-0.060	-0.031	-0.088	-0.020	-0.015
	(0.042)	(0.043)	(0.041)	(0.075)	(0.023)	(0.043)
morelink	0.098***	0.099***	0.098**	0.091**	0.184***	0.129***
	(0.021)	(0.021)	(0.030)	(0.029)	(0.023)	(0.028)
$\ldots \times \mathbb{1}$(second half)	0.063	0.060	0.046	0.040	0.013	0.032
	(0.041)	(0.039)	(0.039)	(0.037)	(0.026)	(0.033)
act	-0.300**	-0.299**	-0.491***	-0.482***	-0.255***	-0.260***
	(0.087)	(0.088)	(0.075)	(0.078)	(0.041)	(0.041)
$\ldots \times \mathbb{1}$(second half)	0.003	0.011	-0.032	-0.035	-0.069	-0.051
	(0.109)	(0.108)	(0.113)	(0.115)	(0.072)	(0.071)
morelink $\times$ act	0.197**	0.209**	0.404***	0.408***	0.224***	0.264***
	(0.070)	(0.070)	(0.082)	(0.078)	(0.037)	(0.029)
$\ldots \times \mathbb{1}$(second half)	0.020	0.014	0.065	0.102	0.085	0.072
	(0.098)	(0.096)	(0.126)	(0.123)	(0.068)	(0.066)
mpay		0.000		0.004**		0.006**
		(0.001)		(0.001)		(0.002)
$\ldots \times \mathbb{1}$(second half)		0.003*		-0.001		-0.001
		(0.001)		(0.002)		(0.002)
turn		0.006***		0.002		0.011***
		(0.001)		(0.002)		(0.003)
$\ldots \times \mathbb{1}$(second half)		-0.002		0.007*		0.000
		(0.002)		(0.003)		(0.002)
Constant	0.875***	0.802***	0.864***	0.793***	0.791***	0.633***
	(0.022)	(0.030)	(0.026)	(0.051)	(0.014)	(0.039)
Individual Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
$\mathbb{P}$(pooling)	0.300	0.090	0.120	0.013	0.148	0.229
Observations	3276	3276	3072	3072	3162	3162
Adj. R^2	0.160	0.169	0.243	0.256	0.219	0.249

Standard errors clustered at the session level in parentheses.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

C Instructions

Welcome. This is an experiment on individual decision making in groups, and you will be paid for your participation in cash at the end of the experiment. The entire experiment will take place through computer terminals, and all interactions between participants will take place through the computers. You will remain anonymous to me and to all the other participants during the entire experiment; the only person who will know your identity is the Lab Manager who is responsible for paying you in the end. Moreover, it is important that you do not talk or in any way try to communicate with other participants during the experiment.

We will start with a brief instruction period. During the instruction period, you will be given a complete description of the experiment and will be shown how to use the computers. You must take a quiz after the instruction period, so it is important that you listen carefully. If you have any questions during the instruction period, raise your hand and your question will be answered so everyone can hear. If any difficulties arise after the experiment has begun, raise your hand, and an experimenter will come and assist you. Please note that you are not being deceived and you will not be deceived: everything I tell you is true.

Your earnings during the experiment are denominated in tokens. Depending on your decisions, you can earn more tokens or lose some tokens. At the end of the experiment, we will count the number of tokens you have earned in all of the matches and you will receive $1.00 for every 4 tokens. You will be paid this amount plus the show-up fee of $5. Different participants may earn different amounts. Everyone will be paid in private and you are under no obligation to tell others how much you earned.

The experiment will consist of 8 matches. In each match, you will be put in a group with 5 other participants in the experiment. Since there are 12 participants in today's session, there will be 2 groups in each match. You are not told the identity of the participants in your group. Your payoff in each match depends only on your decisions, the decisions of the other 5 participants in your group and on chance. What happens in the other group has no effect on your payoff and vice versa. Your decisions are not revealed to participants in the other group.

We will now explain how each match proceeds. At the beginning of the match, the computer randomly assigns each of you to a group consisting of 6 participants. Next, the computer randomly assigns with equal probability a role to each of the participants as "Subject 1", "Subject 2" and so on up to "Subject 6". Then, the match begins.

Each match consists of several turns. At the beginning of each turn, the computer randomly pairs all subjects within each group with one another. We shall call the subject

that you are paired with at each turn as your "Current Partner". Once everyone receives a Current Partner, a turn begins.

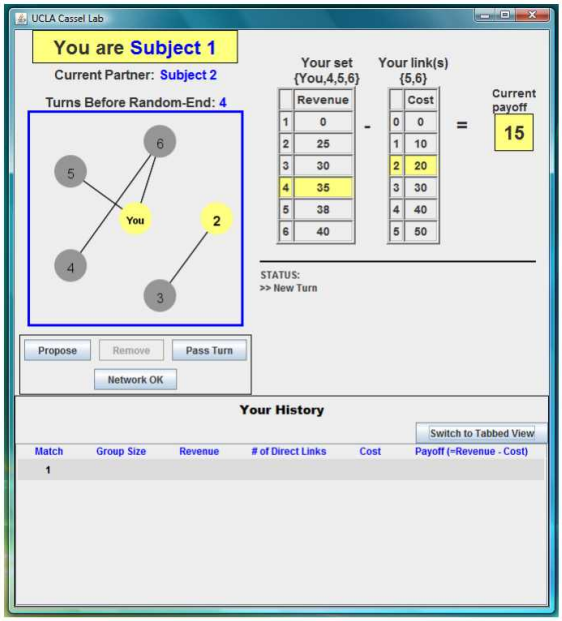


FIGURE 1:
BEGIN A TURN

At the beginning of each turn, you will see a screen similar to that shown here. The top panel provides the information and interface that you will use to interact with other subjects within your group. Meanwhile, the bottom panel lists your payoff history throughout the experiment. Payoff information in each match, including the practice matches, is recorded here.

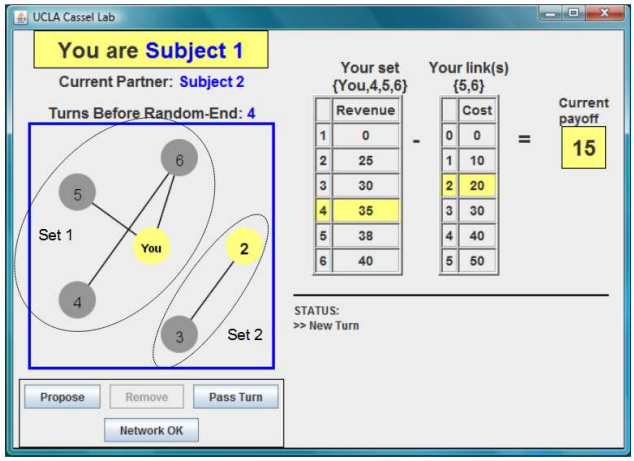


FIGURE 2: SETS, LINKS AND ACTIONS [example 1]

This is the top panel. On the top-left is your role in this match. In this example, you are Subject 1. The computer also informs you of your Current Partner at each turn. In this turn, your Current Partner is Subject 2.

In the middle of the left panel, you will see a network representation of the connections between all subjects in your group. Other subjects in your group are represented by nodes with their role ID numbers. Meanwhile, you are always represented by the center node labeled "YOU". In each turn, the node for your Current Partner is colored YELLOW unlike the rest of the subjects. From the color, you can see here that your Current Partner is Subject 2.

The lines connecting the nodes represent the links between subjects in your group. Everyone in your group sees the same sets of links. In this example, you have direct links to Subjects 5 and 6. Through Subject 6, your are also indirectly connected with Subject 4. Subjects who are either directly or indirectly connected belong in the same "Set". In this example, there are two sets. The first consists of You, Subjects 4, 5, and 6. The second set consists of Subjects 2 and 3.

At each turn, the joint actions of you and your current partner affect how the two of you are linked. You take actions by clicking one of the action buttons below the network representation. Through your actions, you can either propose a link, remove a link, or maintain how you are connected with your partner.

In this first example, since you are not linked to Subject 2, only three actions are available: “Propose”, “Pass Turn”, and “Network OK”. The “Remove” button is not active. Clicking “Propose” lets the computer know that you would like to propose a link with your Current Partner. If your partner does the same, the computer will create a link between you and your partner. Otherwise, no link will be created. In other words, a link is created if and only if BOTH partners propose a link to each other.

If you don't want to link with your Current Partner, you can either click “Pass Turn” or “Network OK”. In either case, a link will not be created. However, notice the difference between the two actions. When you pass a turn, you tell the computer that you want to keep the way you are linked with your current partner in this turn. However, you may still want to change how you are linked with some of the other subjects. So, your buttons will remain active in the next turn

Meanwhile, if you choose “Network OK”, you tell the computer that as long as the network doesn't change, you are happy with the way you are linked with everyone in your group. Therefore, if you click “Network OK”, you won't need to take further actions until the network changes. Your buttons will therefore be inactive. However, these buttons will immediately become active once the decisions of other pairs either break or make a link. If all active subjects choose “Network OK” in the same turn, then the match ends.

The turn ends once everyone in your group has taken an action. The computer then begins a new turn, and you will be randomly assigned a new Current Partner. Please note that since pairs are selected randomly, you may be paired with the same partner in consecutive turns.

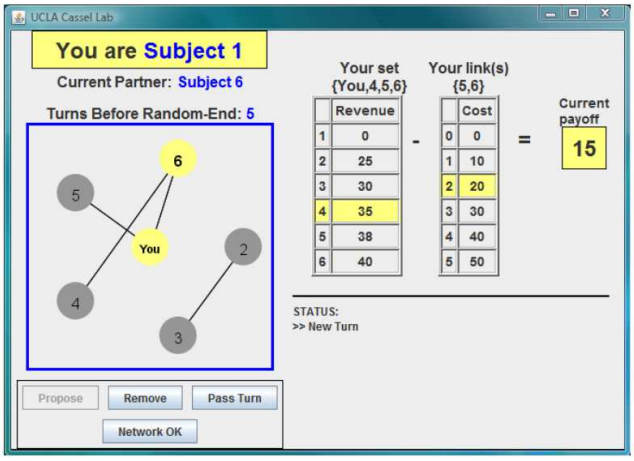


FIGURE 3: SETS, LINKS AND ACTIONS [example 2]

This figure illustrates a new turn in which you are paired with Subject 6. Now, since you are already directly linked with this subject, the three actions available to you are: "Remove", "Pass Turn" and "Network OK". The "Propose" button is deactivated in this turn.

Your link with Subject 6 will remain intact only if BOTH you and Subject 6 don't want to remove it. If at least one subject in the pair wants to remove it, your direct link with your Current Partner will be broken at the end of the the turn. Obviously, the link will also be broken if both subjects in a pair choose to remove it.

In each match, the computer will continue to generate new turns for at least 12 turns unless all subjects choose "Network OK". However, if a match does not end after 12 turns, the match enters the random-end stage. In the random-end stage, at each turn, the computer randomly decides whether it will end the match or generate a new turn. Each time, there is a 20% probability that it will decide to end the match. On average, this implies about 5 additional turns in each match. The number of remaining turns before this random-end stage is displayed above the network representation.

The network representation updates links that are made and broken in real time. You can see changes to the network immediately after each pair makes their decisions within each turn. Similarly, you can also keep track of changes within each turn through the "Status" indicator on the lower right panel. This status indicator resets at each new turn.

We will next discuss about the payoff. Your payoff depends on the size of your set and the number of direct links at the end of the match. Your set size, which is the number of subjects who are either directly or indirectly connected to you, determines your revenue. Meanwhile, your cost is determined by the number of direct links you have.

The right panel provides you with all of the information necessary to calculate your payoff. The table on the left gives you the revenue schedule for different set sizes. Above it, you can see the list of subjects in your set. In this example, your set consists of You and Subjects 4, 5, and 6. Therefore, as part of a set of size 4, your revenue is 35.

Next to the revenue table is the cost schedule for different numbers of direct links. Each direct link incurs a constant cost. In this particular example, the cost for each link is 10 and, therefore, the total cost is 10 times the number of subjects with whom you are directly linked. Above that table, you can see that you are directly linked to Subjects 5 and 6. Since you have two direct links, the current total cost is 20 tokens.

Your current revenue and cost at any stage of the game are highlighted in YELLOW. They are updated in real time as the actions of subjects make and break links within each turn. The rightmost box entitled "Current Payoff" calculates your payoff at each stage of the game. The current payoff is simply the revenue minus cost, which in this case is 15. This payoff information is also updated in real time. Note that the revenue and cost tables may change from match to match.

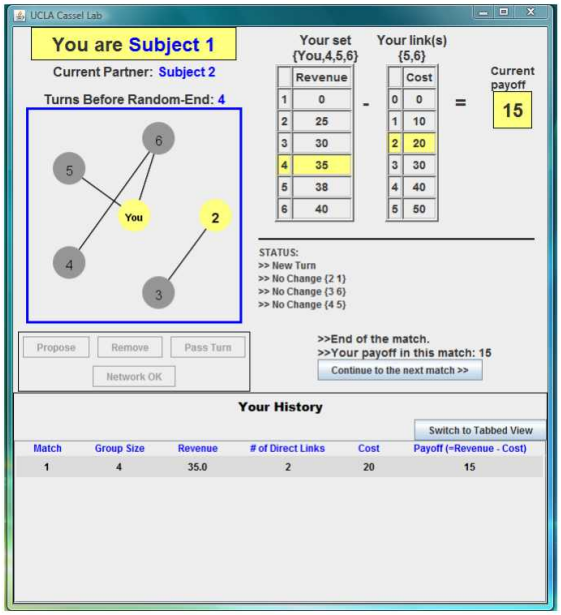


FIGURE 4: END OF A MATCH

This figure illustrates what you will see at the end of a match. Below the status indicator, you will see your payoff for this match. At the end of the match, please click "Continue to the Next Match". In each new match, you will be randomly assigned to a new group. A new match will begin only after all groups have completed their matches. This continues for 8 matches, after which the experiment ends.

At the end of the final match in the experiment, you will see the following screen.

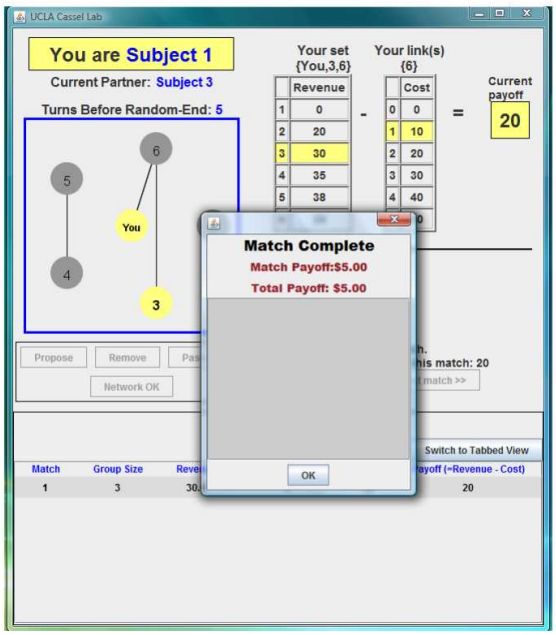
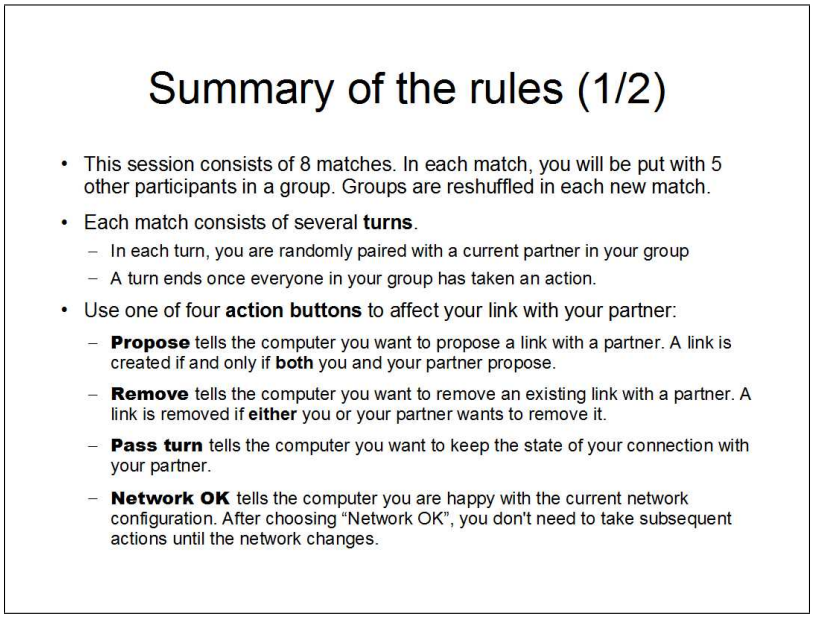
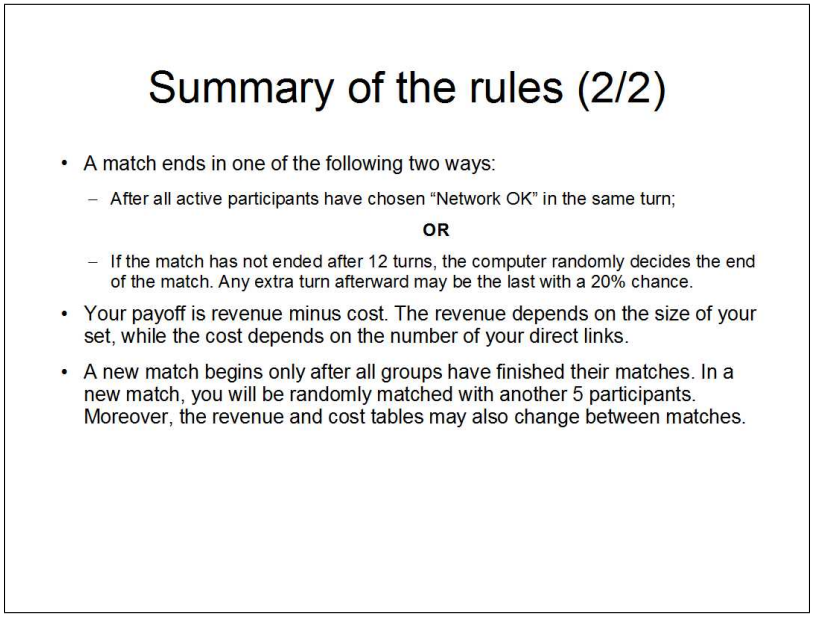


FIGURE 5: END OF A SESSION

This final screen tells you the total payoff that you will receive for this experiment. When you see this screen, don't click OK until you have written down your total payoff on the payoff sheet provided. After you have written down your total payoff, click OK to conclude the session. (*)

The following slides summarize the rules of the experiment:





We will now begin the Practice session and go through two practice matches to familiarize you with the computer interface and the procedures. During these practice matches,

please do not hit any keys until you are asked to. Remember, you are not paid for these matches. At the end of the practice matches you will have to answer some review questions.

Throughout the session, pay attention to the network representation display and status indicators. Also, notice the movements of the yellow highlights on your Revenue and Cost tables, as well as updates to your Current Payoff.

[START GAME]

You have just received a new turn. First, pay attention to your role. If you are Subject 1, 2, or 3, please click "Propose". For Subject 1, 2, or 3, notice a link has just been created between you and your partner if your partner is also Subject 1, 2 or 3.

Now, if you are Subject 4, 5, or 6, please click the "Pass Turn" button. Notice here that a link is created if and only if BOTH partners propose a link. If only one partner proposes a link, no link is created.

You have moved to a new turn. We will now see how the "Network OK" action works. If you are either Subject 5 or 6, please click "Network OK". For the rest of the group, please click "Pass Turn".

You have moved to a new turn. For Subjects 5 or 6, since the network has not changed after you clicked "Network OK", all of your buttons are now inactive. Notice that they will become active following a change in the network.

For others, please check your Current Partner. If your partner is not Subject 5 or 6, click the "Remove" button if it's active or "Propose" otherwise. For Subjects 5 and 6, notice how a change in the network activates your buttons.

If you are not Subject 5 or 6 and your buttons are still active, please click "Pass Turn". If you are Subject 5 or 6 and your buttons are active, please click "Pass Turn". Notice here that if your buttons are inactive due to a "Network OK" action in a previous turn, a change in the network will immediately activate your buttons. In the following, we will do the same exercise for Subjects 1 to 4.

You have moved to a new turn. If you are Subject 3 or 4, please click "Network OK". For the rest of the group, please click "Pass Turn".

You have moved to a new turn. Subjects 3 and 4, notice that your buttons are inactive. If the network changes in this turn, your buttons will become activated.

For all others, check your Current Partner. If your partner is not Subject 3 or 4, click "Remove" if it's active or click "Propose" otherwise. If you are not Subject 3 or 4 and

your buttons are still active, click “Pass Turn”. Now, if you are Subject 3 or 4, please click “Pass Turn”.

You have moved to a new turn. If you are either Subject number 1 or 2, please click “Network OK”. For the rest, please click “Pass Turn”.

You have moved to a new turn. For Subject 1 or 2, your buttons are now inactive. For all others, if your Current Partner is not Subject 1 or 2, click the “Remove” button if it's active, or click “Propose” otherwise. For everyone else who has not taken an action, please click “Pass Turn”.

You have moved to a new turn. Notice from the message above the network display that this is the last turn before the random-end stage. During the paid match, you will have 12 turns before entering this stage. If the match has not ended after 12 turns, the computer will randomly decide the end of the match.

We will now deliberately end the match. If your buttons are active, please click the “Network OK” button. This ends the first practice match. The bottom part of your screen contains a table summarizing the results for all matches you have participated in. This is called the history screen. It will be filled out as the experiment proceeds. Now click “Continue to the Next Match”. We will now begin with the second practice match.

[NEXT MATCH]

You are in a new match. Note here that the revenue and cost tables have changed as they may during the real matches. We'll now examine the behavior of the “Remove” action.

If you are either Subject 2, 4, or 6, please click “Remove”. For Subjects 1, 3, and 5, please click “Pass Turn”. Hence, notice that a link is broken if at least one of the partners chooses to remove it.

You have moved to a new turn. Next, we'll see what will happen if the network changes within the turn in which you click “Network OK”. If you are Subject number 1, 3, or 5, please click the “Network OK” button. For all others, please click your “Remove” button.

You have moved to a new turn. For Subjects 1, 3, or 5 notice that if in the previous turn the network changed after you clicked “Network OK”, your action buttons are active in this turn. If the network did not change after you clicked “Network OK”, your buttons remain inactive. Now, if you are either Subject 2, 4, or 6, click “Network OK”. For all

others, if you haven't taken an action in this turn, please click the "Remove" button if it's active, or "Propose" otherwise.

You have moved to a new turn. Similarly for Subjects 2, 4, and 6, notice that if in the previous turn the network changed after you clicked "Network OK", your buttons are now active. If the network did not change after you clicked "Network OK", your buttons are still inactive. If the network changes in the same turn and after you choose "Network OK", your buttons stay active in the following turn.

We will now end the match. If your buttons are active, please click "Network OK". This ends the second practice match.

*** END OF PRACTICE SESSION ***

The practice matches are over. Please click "Continue to the next match" and complete the quiz. It has 8 questions in two pages. You will move to the next page once everyone in your group has completed the questions in that page correctly. On your table, you will find the screenshots that you will need to answer these questions. Raise your hand if you have any questions.

[WAIT for everyone to finish the quiz]

Are there any questions before we begin with the paid session? We will now begin with the 8 paid matches. Please pull out your dividers. If there are any problems or questions from this point on, raise your hand and an experimenter will come and assist you.

[START MATCH 1]

[After MATCH 8, read:]

This was the last match of the experiment. Now, please write down your ID on the payment sheet. Your ID is located on top of your physical monitor and it began with CASSEL. At this point, if you haven't clicked "Continue to the next match", please do so. Your total payoff is displayed on your screen. Please record this payoff in the earned column of your sheet and sign it. Once you have written it down, please click OK.

Your Total Payoff will be this amount rounded up to the nearest dollar plus the show-up fee of $5. We will pay each of you in private in the next room. Remember you are under no obligation to reveal your earnings to the other subjects.

If you are done, please line up behind the yellow line until the lab manager calls you to be paid. Do not converse with the other subjects or use your cell phone. Thank you for your cooperation.